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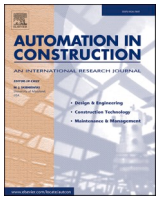

Digital twin of advanced geological conditions in rock tunnel construction

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ABSTRACT

Geological uncertainty ahead of the tunnel face presents significant challenges to the safety and efficiency of tunnel construction. To address this issue, this paper explores the feasibility of using a digital twin approach to enhance geological modeling and rock classification. The proposed method integrates multi-source geological data to construct a multi-dimensional geological virtual model, and applies a fuzzy cloud probability model to quantify uncertainty, enabling intelligent classification of surrounding rock. Results from tests on five rock tunnel projects using the iS3 platform show that the overall rock classification accuracy of the method is 92.95 %. These findings clearly demonstrate that the digital twin approach significantly improves geological transparency, risk management, and decision-making efficiency in complex rock tunnel environments. Future research will focus on exploring the application of this method in soft soils, underwater tunnels, and other broader geological conditions.

1. Introduction

Rock tunnel construction faces complex and unpredictable geological conditions that can jeopardize construction safety and affect project efficiency [1–7]. Advanced geological forecasting plays a key role in detecting adverse conditions ahead of the tunnel face [8–11]. However, traditional methods mainly rely on limited data sources and the subjective judgments of geological engineers, which often fail to fully capture the inherent uncertainty and complexity of rock mass behavior [12–15]. To address these issues, there is an urgent need for new approaches that integrate multi-source data, enabling a comprehensive and real-time interpretation of geological conditions. In recent years, digital twin technology has emerged as a promising solution in the field of tunnel construction. This technology integrates multi-source heterogeneous data to construct high-precision virtual geological models and employs intelligent algorithms to enable real-time interactions between physical entities and their digital twins, thereby providing engineering managers with intuitive, dynamic monitoring and early warning capabilities. However, most current geological digital twin models lack rigorous mathematical descriptions of the randomness and fuzziness inherent in rock mass parameters and do not establish a complete system for data acquisition, model updating, and decision feedback [16–18]. This limitation restricts their effectiveness in real-time construction applications [19].

Based on the aforementioned background and technological status, this paper focuses on the following research question: How can a closed-loop digital twin system for Data-Model-Decision be constructed to achieve real-time and accurate decision of advanced geological conditions within a 30-m range ahead of the tunnel face during excavation? The “Data-Model-Decision” is a concept designed to progress step by step, starting from data collection, followed by virtual model construction, and ultimately supporting decision-making, forming a closed-loop system.

To address this issue, the paper proposes a three-step digital twin approach: (1) Multi-source information collection and database construction: Geological information, such as water content, lithology, and integrity, is extracted using image processing and other methods. This information is aligned in three-dimensional space and transformed into consistent point cloud data structures stored in a database. (2) Advanced geological virtual model construction: Point cloud models ahead of the tunnel face are generated based on the geometric characteristics of the tunnel face. Geological attributes are assigned to the point cloud model using the KNN interpolation based on information from the geological database. (3) Intelligent decision-making for surrounding rock classification: The geological attributes of the constructed point cloud models are used as input to a fuzzy cloud probability model to quantify uncertainties in the geological information, allowing for the calculation of rock classification probabilities for each meter ahead of the tunnel face. The classification is determined based on the maximum probability. The “advanced geological conditions” referred to in this paper pertain to the geological conditions within a 30-m range ahead of the tunnel face

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Nomenclature

r_x, r_y, r_z	Coordinates of the sample to be interpolated
o_x, o_y, o_z	Coordinates of the sample from the database
$d(r, o_i)$	The distance between point r and point o_i
K	Number of neighbors in KNN interpolation
y_j	Geological attributes of sample point j
$\hat{y}$	Mean of geological attributes
Ex	Expected value of cloud model
En	Entropy value of cloud model
He	Hyper-entropy value of cloud model
Enn	Random number generated with a normal distribution, where En serves as the mean and He^2 as the variance
x_n	Random number generated with a normal distribution, where Ex is the mean and Enn^2 is the variance
z_n	Deviation of x_n
Lx^{max}, Lx^{min}	Maxmum and minimum values of the interval
f	Normal distribution membership function
v	Boundary value of the normal distribution function
N_i	The number of cloud drops in i region
P_i	Surrounding rock grade probability
WI	Water content index
WK	Groundwater correction factor
BQ	Basic quality indicator of the rock mass
$[BQ]$	Engineering rock mass quality indicator
Kv	Integrity index of the surrounding rock
Rc	Uniaxial compressive strength of the rock
WP_i	Water-richness level probability

during tunnel construction. This method features dynamic data collection, multidimensional virtual mapping of physical entities, and decision support that quantifies uncertainty, all of which are key characteristics of digital twin. Compared to traditional advanced geological forecasting methods that rely mainly on a single data source and engineer experience, and thus exhibit significant shortcomings in handling the uncertainties of complex geological conditions, the digital twin method proposed in this paper is considerably more advanced. By constructing a multidimensional virtual model that integrates water content, lithology, and integrity information, and by employing a fuzzy cloud probability model to quantify uncertainties, this method achieves real time data collection, dynamic modeling and intelligent decision making, fundamentally overcoming the deficiencies of traditional approaches in data integration and real time decision making. The Data-Model-Decision process for digital twin of advanced geological conditions is shown in Fig. 1.

2. Related work

This section mainly introduces the research background and current state of studies relevant to this paper, while also discussing the existing gaps in the current research. It is organized into three main aspects: Advanced geological forecasting in tunnel construction, Digital twin in tunnel construction, and Cloud model in tunnel engineering applications.

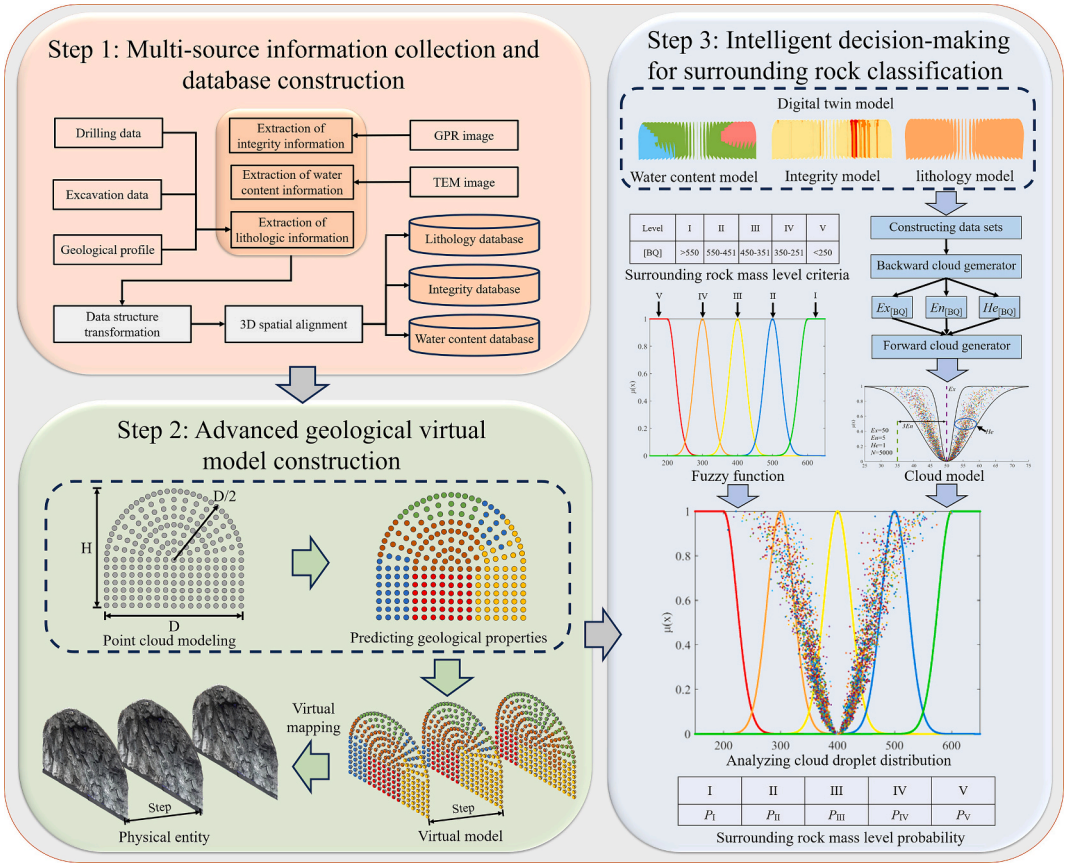


Fig. 1. Data-Model-Decision process for digital twin of advanced geological conditions.

2.1. Advanced geological forecasting in tunnel construction

Advanced geological forecasting in tunnel construction aims to identify the geological conditions ahead of the tunnel face in advance, effectively preventing accidents caused by adverse geological factors [20,21]. Traditional advanced geological forecasting methods, such as GPR, TSP, TEM, and while-drilling, use different physical principles to obtain information about the physical properties of the geology ahead of the tunnel face [22–24]. Under normal conditions, the effective detection ranges are as follows: GPR (15-30 m), TSP (100-150 m), TEM (30-60 m), and while-drilling measurements (3-15 m). These methods provide foundational data for assessing geological conditions, but due to the complexity of geology, a single method often fails to capture the uncertainty and complexity of the geology [25]. Currently, tunnel construction sites mainly rely on geological engineers to subjectively assess the geological conditions ahead of the tunnel face based on information from advanced geological forecasting. There is a lack of a method that can automatically make decisions regarding the geological conditions ahead of the tunnel face based on the data provided by advanced geological forecasting. Therefore, researchers have begun to explore the integration of multi-source data to make tunnel geology more transparent, such as combining GPR with drilling to complement the advantages of different methods. Additionally, in recent years, computer technology has been increasingly applied to geological forecasting, utilizing data-driven methods to extract valuable patterns from historical data and improve the accuracy of geological condition predictions [26]. Wang et al. proposed a semi-supervised learning model based on while-drilling data, which combines dense convolutional autoencoders and hybrid deep neural networks. This approach significantly improved the accuracy and stability of surrounding rock discontinuity detection [27]. Xue et al. adopted a novel spatiotemporal sequence approach to interpret the continuously exposed rock mass during tunnel excavation. They proposed a spatially aware recurrent neural network that uses historical fracture maps as input to predict the unexcavated sections [28]. Hansen et al. proposed a machine learning method that uses while-drilling data to predict the rock type ahead of the tunnel face [29]. These new methods have significantly improved the accuracy and real-time performance of advanced geological forecasting. However, challenges remain in effectively integrating multiple data sources and handling uncertainties.

2.2. Digital twin in tunnel construction

Digital twin technology is an emerging technology that has rapidly developed in the engineering field in recent years. By creating a virtual replica of a physical entity, it connects real-time data from the physical world with virtual models, enabling dynamic reflection and prediction of physical processes [30–33]. In tunnel construction, digital twin technology can integrate geological, structural, and construction data in real time to create a dynamic virtual model of the construction site. This allows engineers to identify potential risks in a virtual environment and optimize construction plans [34–36].

The advantage of digital twin technology lies in its ability to simulate and predict the constantly changing geological conditions during tunnel construction with high precision, while also offering powerful data analysis and visualization capabilities [37]. Zhang et al. installed Internet of Things sensors on TBM to transfer the TBM's actual operating status from the physical object to the digital model [38]. Ye et al. proposed a digital twin-based multi-information intelligent early warning and safety management platform. This platform sets up communication lines within the tunnel using multifunctional base stations to automatically collect and transmit multi-source data. It then uses this data to virtually simulate dynamic construction conditions and surrounding rock behaviors [39]. Ling et al. integrated digital twin technology into the implementation of rock tunnel Just-In-Time design and proposed a scene-entity-relationship-event-control modeling approach. This new method serves as a cross-domain sharing and reuse model for digital twin applications [40]. Wu et al. proposed a multi-level voxel representation method based on linear segmentation, which efficiently updates and expresses the complex features of the tunnel geological environment, providing real-time dynamic support for digital twin applications [41]. However, the construction of digital twins relies on the efficient acquisition and integration of multi-source data, which demands high data processing capabilities and modeling accuracy. Additionally, the geological environment in rock tunnel construction is highly complex and uncertain, making it a challenge to quantify these uncertainties and make intelligent decisions based on geological conditions in the context of digital twin applications.

2.3. Cloud model in tunnel engineering applications

The cloud model, a mathematical tool for addressing uncertainty and fuzziness, has been widely used in tunnel engineering applications [42]. The core advantage of the cloud model lies in its ability to convert between qualitative concepts and quantitative descriptions, making it particularly well-suited for capturing the variability and uncertainty of geological conditions [43]. In tunnel construction, the geological conditions are often influenced by multiple factors, such as lithology, water content, and structural integrity. The interactions between these factors are complex and characterized by high uncertainty [44–47].

The cloud model uses membership functions and probability distributions to transform the multi-dimensional features of surrounding rock conditions into a comprehensive probability cloud map, providing engineers with an intuitive basis for risk assessment. Compared to traditional single-source risk assessment methods, the advantage of the cloud model lies in its ability to effectively quantify both fuzziness and randomness, thereby enhancing the accuracy and dynamic nature of geological condition evaluation. Sun et al. proposed a dynamic risk assessment system that combines a two-dimensional cloud model for risk early warning in tunnel construction. This approach effectively handles the uncertainties in geological conditions and provides a reliable decision-making basis for the construction process [48]. Chen et al. proposed a hybrid method that combines an improved cloud model with a Bayesian network for dynamic safety analysis and assessment of adjacent buildings impacted by tunnel construction [49]. Tian et al. integrated resilience theory with a combined weighted-cloud model to assess the flood resilience of metro stations [50]. Zhu et al. proposed a cloud model-based random forest method to address inconsistencies in mapping monitoring data to the health rate of metro tunnels [51]. In this study, the cloud model is improved by incorporating fuzzy mathematics theory, resulting in the development of a fuzzy cloud probability model. This model is used to quantify the uncertainty of the constructed geological digital twins and make intelligent decisions regarding surrounding rock classification based on probabilities, providing crucial support for safety decision-making in rock tunnel construction.

3. Methodology

This section primarily presents the theoretical framework of the proposed digital twin method for advanced geological conditions in tunnel construction. First, the section on Multi-source information collection and database construction describes how a multi-dimensional geological database is established by gathering data from various sources. Next, in section 3.2 Advanced geological virtual model, a point cloud model representing the geological conditions ahead of the tunnel face is created based on the tunnel's geometric features. Geological attributes are then assigned to the point cloud using the KNN algorithm, leveraging the geological database to construct a multi-dimensional virtual model of the advanced geological conditions. Section 3.3 focuses on the theoretical approach of the fuzzy cloud probability model, which uses the point cloud data from the virtual model as input to quantify uncertainties in the geological data, enabling intelligent

decision-making for surrounding rock classification.

3.1. Multi-source information collection and database construction

This section focuses on extracting geological information from various sources, such as GPR, TEM, boreholes, geological profile map, and tunnel excavation exposure, using techniques like image processing. The extracted data is then converted into a unified point cloud data structure and spatially registered in three dimensions. This multi-source geological information is stored in a database, creating a dynamic multi-source information system that can be updated as new geological data becomes available.

3.1.1. Lithology information database construction

The construction of the lithology database relies primarily on borehole data and the lithological information exposed during tunnel excavation. Borehole data provides stratigraphic information at various depths along the tunnel. To enhance the spatial resolution of this data, line interpolation is used to convert the borehole data into three-dimensional point cloud data [52,53]. Specifically, borehole depth data are resampled at a fixed step size (in this study, a step size of 1 m was adopted), converting the lithology information at each sampling depth into point cloud data points with precise three-dimensional coordinates. In addition, the geological information exposed during actual tunnel excavation serves as an essential supplementary data source for the lithology database. As excavation progresses, the lithological conditions exposed at the tunnel face are recorded on-site and spatially positioned in three dimensions. Subsequently, these data are integrated with the existing borehole data and transformed into three-dimensional point cloud data using the same linear interpolation approach, maintaining a consistent 1-m step interval. These newly incorporated data continuously enrich the lithology database, enabling it to be dynamically updated and optimized throughout the construction process, thus ensuring accurate, real-time representation of geological conditions ahead of the tunnel face.

Geological profile maps constitute another critical data source. These maps, drawn by geological engineers, illustrate lithological distributions along the tunnel axis and embody the engineers' understanding of geological structures [54]. To effectively convert two-dimensional geological profile data into three-dimensional point cloud data, this study adopts a digital processing workflow: (1) RGB values of each pixel in the geological profile maps are first extracted; (2) Lithology information is assigned to each pixel by establishing a mapping relationship between RGB values and lithological categories; (3) Based on spatial scaling relationships derived from geological profile maps, each pixel's lithological information is accurately registered in three-dimensional coordinates; (4) The lithological information, along with corresponding spatial coordinate points, are formatted into structured datasets and stored to construct the comprehensive lithology database.

Finally, borehole data, tunnel excavation exposure data, and geological profile map data are integrated into a unified three-dimensional point cloud database. After spatial registration and data consistency processing, the consolidated database accurately reflects the geological features along the tunnel alignment. The process is illustrated in Fig. 2.

3.1.2. Integrity information database construction

The extraction of surrounding rock integrity information ahead of the tunnel face is a critical step in tunnel engineering, providing essential data for safety assessments during tunnel construction. GPR is a predictive geological method based on electromagnetic waves [55]. The degree of fluctuation in the GPR electromagnetic waves ahead of the tunnel face can partially reflect the level of integrity in the surrounding rock. In this study, image processing techniques are used to extract the integrity information of the surrounding rock from the GPR data. This information is then converted into three-dimensional point cloud data, providing support for subsequent analysis and modeling.

The specific steps for processing GPR images are as follows: (1) Image preprocessing: To effectively extract key information that reflects the integrity of the surrounding rock, this study applies morphological operations. The process starts with erosion, followed by dilation to remove small and insignificant linear noise features in the image, while preserving obvious fluctuation regions that reflect rock mass fragmentation. A 4 × 4 square structural element is chosen during the morphological operation, as its size is suitable for processing fine lines and small noise points, while retaining the shape of the main black areas. Through this process, the original GPR electromagnetic wave image is transformed into a binary image that intuitively reflects the degree of surrounding rock fragmentation. (2) Fragmentation area identification: However, the identified main fluctuation areas still appear wavy. In actual engineering, the fragmentation of the surrounding rock typically manifests as irregular block-like fragments. Therefore, elliptical fitting is further applied to the contour of each main fluctuation area to generate elliptical regions that more realistically

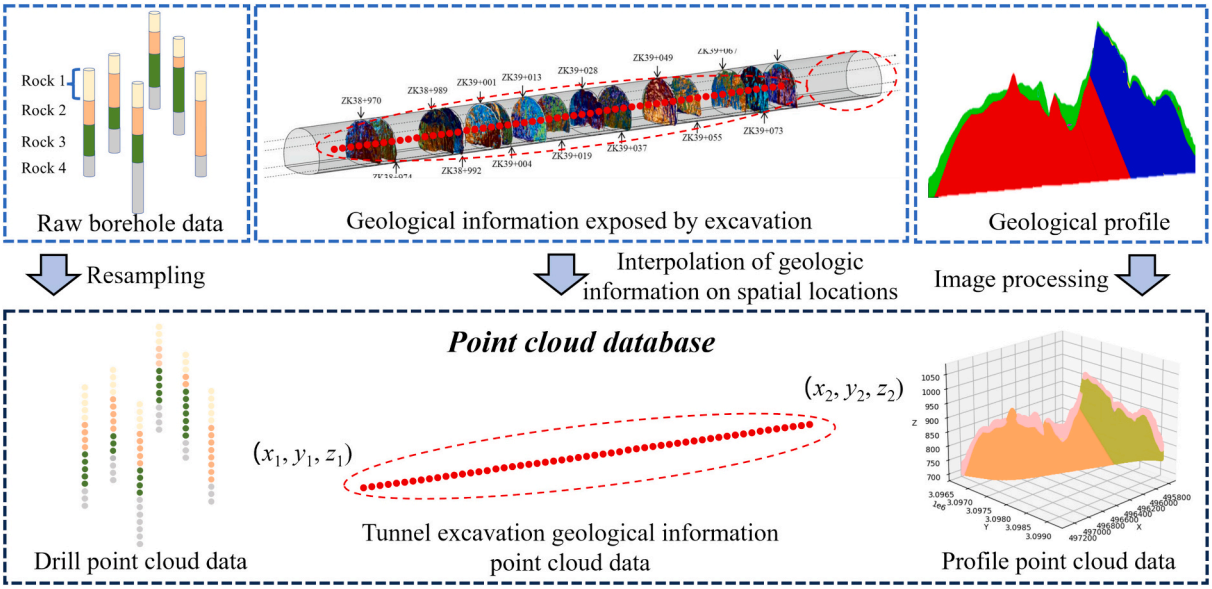


Fig. 2. Lithology database construction process.

represent the rock mass fragmentation. In this operation, areas greater than 100 are selected as the main fluctuation areas, and the axes of the fitted ellipses are enlarged by 1.2 times and filled. (3) Merging two GPR interpretation results: Each GPR scan is performed twice independently, generating two electromagnetic wave images. To ensure the reliability of the results, this study cross-validates the two images and merges the detected rock mass fragmentation areas for analysis. The areas detected in both tests are defined as overlapping regions (marked in red), areas detected in only one test are defined as non-overlapping regions (marked in blue), and the remaining regions with no fluctuation are defined as stable regions (marked in green). Different integrity indicators are assigned to the red, blue, and green regions based on the variation in rock mass fragmentation at the tunnel face. (4) Three-dimensional spatial registration of integrity information: Finally, the extracted surrounding rock integrity information is registered with the actual three-dimensional spatial coordinates in the form of a binary image. This process determines a reasonable spatial coordinate range and uses linear interpolation to map the extracted information from the image to the actual three-dimensional coordinates, forming a 3D point cloud database of surrounding rock integrity. The specific process is shown in Fig. 3.

3.1.3. Water content information database construction

The extraction of geological water content information is a key component in rock tunnel construction and geological forecasting, providing essential data for groundwater management and the safety of the rock tunnel [56]. In this study, the image data obtained from TEM scanning is processed to extract the water content information of the geology, which is then converted into three-dimensional point cloud data for subsequent analysis.

The specific process steps for processing TEM images and constructing a water content information database are as follows: (1) Image preprocessing: During the image processing stage, the data is converted into NumPy arrays for numerical computation and analysis. To ensure the extraction of valid color information, a color distance threshold is set to filter out pixels close to white, thereby avoiding noise interference during analysis. In this study, the color distance threshold is set to 90, which effectively distinguishes near-white pixels from other colored pixels. (2) Water content information extraction: Based on the definition of fissure water pressure for different water-bearing rock types in the standard for engineering classification of rock mass, the resistivity ranges obtained from TEM are assigned different water content indices: resistivity values of 0–800 are assigned a water content index of 1.5 (represented by red); 800–1600 are assigned a water content index of 1 (represented by yellow); 1600–2400 are assigned a water content index of 0.2 (represented by green); and resistivity values greater than 2400 are assigned a water content index of 0.01 (represented by blue). In practical engineering applications, the water content indices corresponding to different resistivity ranges should be adjusted according to the specific conditions of the project and the characteristics of the instrument. After establishing the mapping rules for color-resistivity-water content, all valid pixels in the image are traversed, and their RGB values are compared to determine the corresponding water content index. For each identified water-bearing pixel, its coordinates within the image and the corresponding water content index are recorded and stored in a list or data table, forming the preliminary results of water content information extraction. (3) Three-dimensional spatial registration of water content information: After extracting the water content information, three-dimensional spatial registration is performed. Based on the actual acquisition position of TEM and the proportional relationship between the image and the detection range, the pixel coordinates in the image are mapped to the actual X, Y, and Z spatial coordinates, forming preliminary three-dimensional coordinate data. This information includes the spatial location of each pixel and its corresponding water content index. These data are stored in a structured format, constructing the water content information database. The process of constructing the water content information database is illustrated in Fig. 4.

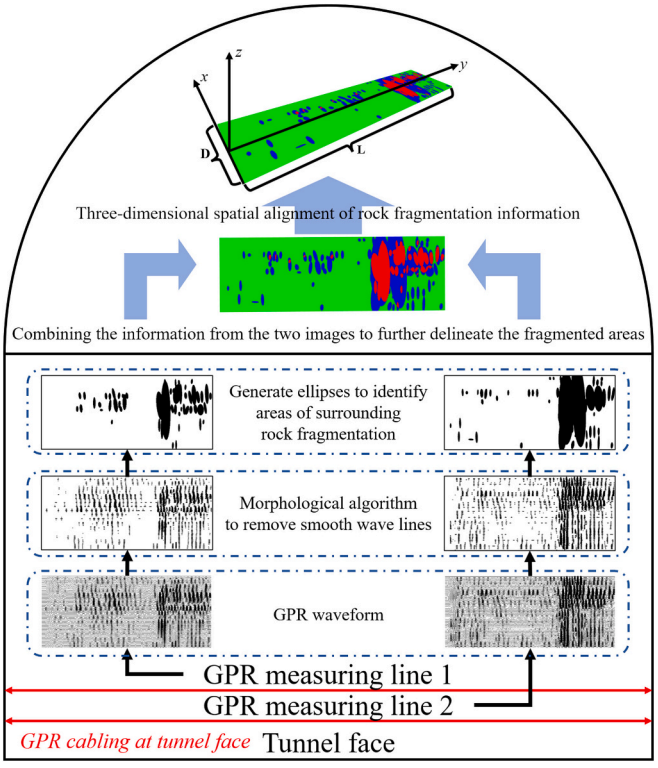


Fig. 3. Process of integrity information extraction and spatial registration construction.

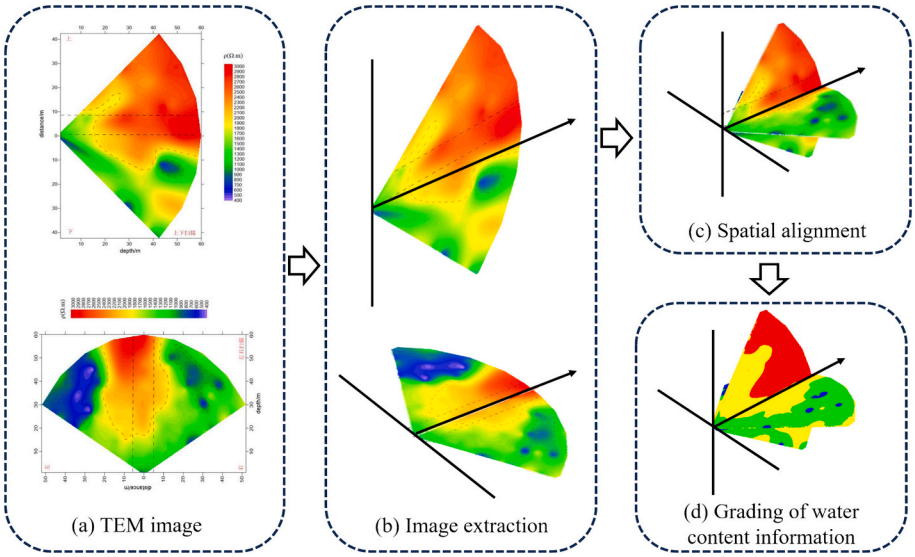


Fig. 4. Process of geological water content information extraction and spatial registration.

3.2. Advanced geological virtual model of tunnel

Section 3.2 primarily discusses the process of constructing the advanced geological virtual model for the tunnel. This section first introduces the key mathematical theory behind the KNN interpolation. It then explains how the KNN algorithm is applied to construct the advanced geological virtual model using a point cloud representation.

3.2.1. KNN interpolation

The KNN algorithm is a widely used interpolation method. A detailed mathematical derivation of this interpolation technique can be found in the study by Chen et al. and Yang et al. [23,57]. The KNN algorithm offers superior interpretability and low computational complexity, making it highly suitable for integration with digital twin technology for rapid geological condition analysis and feedback. However, currently popular machine learning algorithms are not suitable for constructing geological digital twin models. This is because these algorithms typically require large numbers of target variables for validation, while verifying geological conditions obtained through advanced geological forecasting relies on the geological information exposed during tunnel excavation. Within the prediction range of a single geological forecast, typically only geological information from one or two exposed tunnel faces can be collected, which is insufficient for machine learning methods to operate. Furthermore, the "black-box" nature of machine learning algorithms significantly limits their applicability on construction sites where stringent safety standards and transparency are essential. Therefore, this study uses KNN interpolation to construct a geological virtual model. The basic idea is to calculate the distance between the sample to be predicted and the samples in the training set, identify the K nearest neighbors, and make predictions based on the attributes of these neighbors. The workflow of the KNN interpolation includes the following steps:

(1) Distance Calculation**:** For each sample point r to be interpolated, the distance between r and each sample point x_i is calculated first. Common distance metrics include Euclidean distance and Manhattan distance. In this study, Euclidean distance is used, defined by the following formula:

$$d(r, o_i) = \sqrt{(r_x - o_{i,x})^2 + (r_y - o_{i,y})^2 + (r_z - o_{i,z})^2} \tag{1}$$

where r_x, r_y, r_z are the coordinates of the sample to be interpolated, and o_x, o_y, o_z are the coordinates of the sample from the database [23].

(2) Neighbor selection**:** Based on the calculated distances, select the K sample points from the database that are closest to the sample to be interpolated. Assume the states of these K neighbors are y_1, y_2, …, y_K.

(3) Interpolation: For regression problems, KNN uses the average value of the neighbors as the interpolated value:

$$\hat{y} = \frac{1}{K}\sum_{j=1}^{K} y_j \tag{2}$$

For classification problems, KNN applies a majority voting rule, selecting the most frequently occurring class among the neighbors as the interpolated result. In this study, the KNN algorithm is used for constructing virtual model of geology through the following steps: First, features (X, Y, Z coordinates) and target variables (geological conditions) are extracted from a dataset containing spatial coordinate points and geological states. Next, a K-Nearest Neighbors model is initialized, with the number of neighbors set to 100. The number of neighbors is set to 100 because the 3D point cloud model within the 30 m ahead-of-face region is extremely dense—averaging thousands of samples per meter. Selecting too few neighbors renders the interpolation overly sensitive to local noise, causing discontinuities, while selecting too many produces overly smooth results that obscure critical geological anomalies. Extensive experiments indicate that classification accuracy remains most stable when K ranges from 80 to 120. Consequently, $K = 100$ is chosen as the midpoint to balance smoothness with the ability to resolve important spatial variations. The geological properties of the unknown point are interpolated based on the geological properties of the 100 known points surrounding it. Lastly, the point cloud data is color-coded according to the interpolated results to visualize different geological conditions. Through the effective application of the KNN interpolation, this study offers a solid scientific basis for the dynamic monitoring and analysis of advanced geological conditions in tunnel projects.

3.2.2. Point cloud model construction

In tunnel construction, building an accurate point cloud model is a crucial step for achieving dynamic virtual mapping of the geological conditions ahead of the tunnel face [58,59]. In this study, databases for lithology, integrity, and water content information are constructed separately. These databases, comprised of point cloud data, provide a representation of the geological conditions ahead of the tunnel face to a certain extent.

After constructing the point cloud data, the KNN algorithm is applied to interpolate the geological properties of the spatial coordinate points in the point cloud. KNN is a distance-based algorithm that estimates the

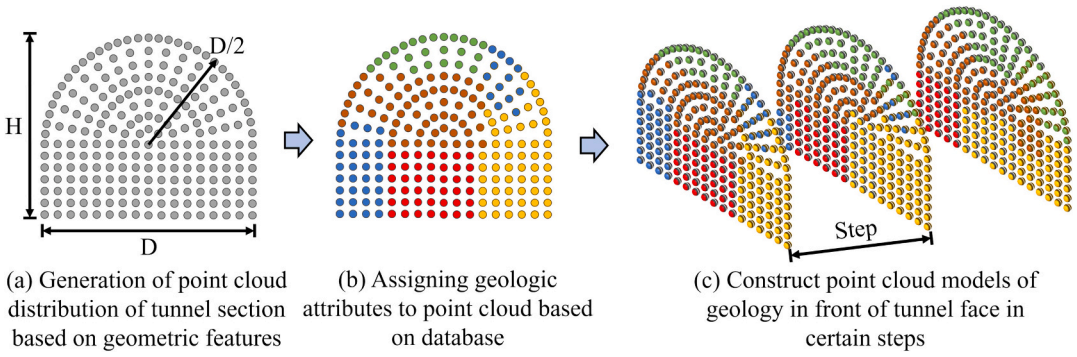


Fig. 5. Point cloud model construction workflow.

state of a point by selecting the K nearest neighbors. In this study, each spatial point in the point cloud contains three-dimensional coordinates (X, Y, Z) along with information about the geological condition. For each new point ahead of the tunnel face, the KNN model calculates its distance to other known points in the database, selects the K closest neighbors, and uses their condition values to estimate the lithology, integrity, and water content of the new point.

In the implementation process, a dense three-dimensional point cloud is first generated to cover the area ahead of the tunnel face. This area consists of a semicircular region representing the tunnel crown cross-section and a rectangular region representing the tunnel bottom. A series of points are generated along the tunnel direction (Y-axis) at fixed intervals, and densely distributed in both the radial and angular directions, creating a point cloud model ahead of the tunnel face. Through these steps, the constructed point cloud cross-sections represent the geometry of the tunnel face and extend along the tunnel direction at a fixed step size. Next, for each point, the KNN algorithm calculates its distance to the existing points in the database and selects the K nearest neighbors. By computing a weighted average of the condition values of these neighbors, the algorithm assigns the lithology, integrity, or water content status to each point in the point cloud model.

To visually present the prediction results, the geological condition values are visualized using color coding, with different values represented by different colors. By mapping these colors onto the three-dimensional point cloud model, the study provides an intuitive display of the geological conditions ahead of the tunnel face, offering a clear and quantifiable basis for construction decision-making. The workflow for constructing the point cloud model is illustrated in Fig. 5.

3.3. Surrounding rock classification intelligent decision-making

Section 3.3 primarily explains how to use the proposed fuzzy cloud probability model for intelligent decision-making in surrounding rock classification. This section first introduces the fundamental mathematical theory behind the fuzzy cloud probability model. Building on this theory, it then integrates the calculation methods for surrounding rock classification, creating a process that uses the multi-dimensional geological virtual model as input, quantifies geological uncertainties, and ultimately makes intelligent decisions regarding surrounding rock classification.

3.3.1. Fuzzy cloud probability model

The cloud model has unique advantages in handling uncertainty transformations, allowing for the quantitative analysis of both the randomness and fuzziness of indicators. A detailed mathematical derivation of cloud model theory can be found in the study by Li et al. [60]. The cloud model is primarily characterized by three parameters: *Ex*, *En*, and *He*. The process of calculating these characteristic values is called the cloud backward generator, and the calculation formulas are as follows:

$$\begin{cases} Ex = \dfrac{1}{q}\sum_{i=1}^{q} x_i \\ S^2 = \dfrac{1}{q-1}\sum_{i=1}^{q}(x_i - Ex)^2 \\ En = \sqrt{\dfrac{\pi}{2}} \times \dfrac{1}{q}\sum_{i=1}^{q}|x_i - Ex| \\ He = \sqrt{|S^2 - En^2|} \end{cases} \tag{3}$$

where *Ex* is the expected value, representing the center of the cloud drop distribution in the domain space and serving as the representative point of the qualitative concept. *En* is the entropy value, which indicates the uncertainty of the qualitative concept, reflecting the level of acceptance of this concept. A smaller entropy value indicates lower uncertainty in the data and more reliable results. *He*, known as the hyper-entropy, represents the entropy of the entropy value and indicates the degree of dispersion of the cloud drops when visualized. The smaller *He*, the less dispersed the cloud drops are [60].

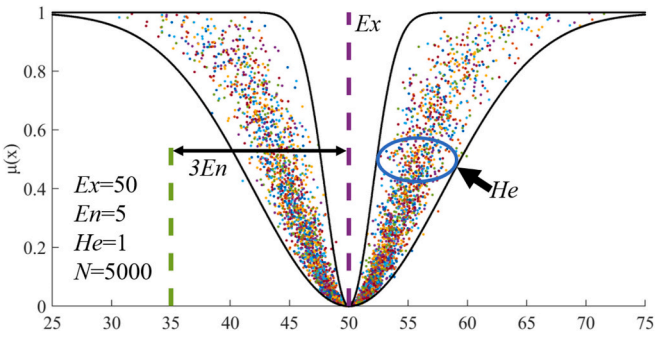


Fig. 6. Cloud model.

The process of simulating cloud drops based on the characteristic values of the cloud model is called the cloud forward generator. This process is primarily accomplished using the membership function of the cloud model. In this study, deviation is used in place of the original membership function. As the cloud drops move away from the expected center, the deviation continuously increases. The formula for the deviation function is as follows:

$$z_n = 1\text{-}e^{-\frac{(x_n-Ex)^2}{2Enn^2}} \tag{4}$$

In the formula, Enn is a random number generated with a normal distribution, where En serves as the mean and He^2 as the variance. Similarly, x_n is a random number generated with a normal distribution, where Ex is the mean and Enn^2 is the variance [48]. The cloud diagram of the cloud model is shown in Fig. 6.

Fuzzy mathematics provides a mathematical tool for handling uncertainty and fuzziness. The core concept is the membership function, which quantifies the degree of membership of an element to a fuzzy set, with values typically ranging from [0,1]. Unlike traditional binary logic (where an element either belongs to a set or does not), fuzzy mathematics allows elements to belong to a set to varying degrees, reflecting the complexity of the real world. In this paper, a method based on constructing fuzzy mathematical normal distribution membership functions is used to create quantized regions of different levels. The formula is as follows:

$$\begin{cases} A_i = \dfrac{Lx_i^{\max} + Lx_i^{\min}}{2} \\ C_i = \dfrac{Lx_i^{\max} - Lx_i^{\min}}{2} \\ \sigma x_i^2 = \dfrac{{C_i}^2}{-2\ln(v)} \end{cases} \tag{5}$$

$$f_i = e^{-\frac{(x-A_i)^2}{2\sigma x_i^2}} \tag{6}$$

In the formula, $Lx^{\max}$ represents the maximum value of the interval, $Lx^{\min}$ represents the minimum value of the interval, i denotes the membership level, f is the normal distribution membership function, and v is the boundary value of the normal distribution function, with v set to 0.1 [48].

By analyzing the distribution of cloud drops in different level regions, the number of cloud drops in each region N_i is calculated. These quantities are then normalized to obtain the probability P_i for each surrounding rock condition. The comprehensive condition of the surrounding rock along each meter is determined based on the highest probability among the surrounding rock condition probabilities.

3.3.2. Quantitative calculation of surrounding rock classification probability

This section focuses on the intelligent decision-making of surrounding rock classification for each meter ahead of the tunnel face using the three constructed point cloud digital twin models. The surrounding rock classification primarily depends on factors such as rock strength, rock integrity, and water content. Due to the significant uncertainty inherent in the constructed point cloud models, this study proposes a fuzzy cloud probability model to quantify the uncertainty in the geological information of the point cloud, enabling intelligent decisions for surrounding rock classification.

First, the comprehensive water content condition of the surrounding rock for each meter ahead of the tunnel face is determined. This process classifies the comprehensive water content of the surrounding rock into three levels based on the water content index (WI): Level I ($WI < 0.1$), Level II ($0.1 < WI < 0.5$), and Level III ($WI > 0.5$). By applying these water content classification intervals to Eqs. (5) and (6), the corresponding membership functions are derived as follows:

$$f_{w,\mathrm{I}} = \begin{cases} 1 & x < 0.05 \\ e^{\left(\frac{x-\frac{0.1+0}{2}}{\frac{0.1-0}{2}}\right)^{2\ln(0.1)}} & x \geq 0.05 \end{cases} \tag{7}$$

Table 1
Groundwater correction factor standard.

Water-rich state of the surrounding rock	Ex_{BQ}				
	>550	550–451	450–351	350–251	≤250
I	0	0	0–0.1	0.2–0.3	0.4–0.6
II	0–0.1	0.1–0.2	0.2–0.3	0.4–0.6	0.7–0.9
III	0.1–0.2	0.2–0.3	0.4–0.6	0.7–0.9	1

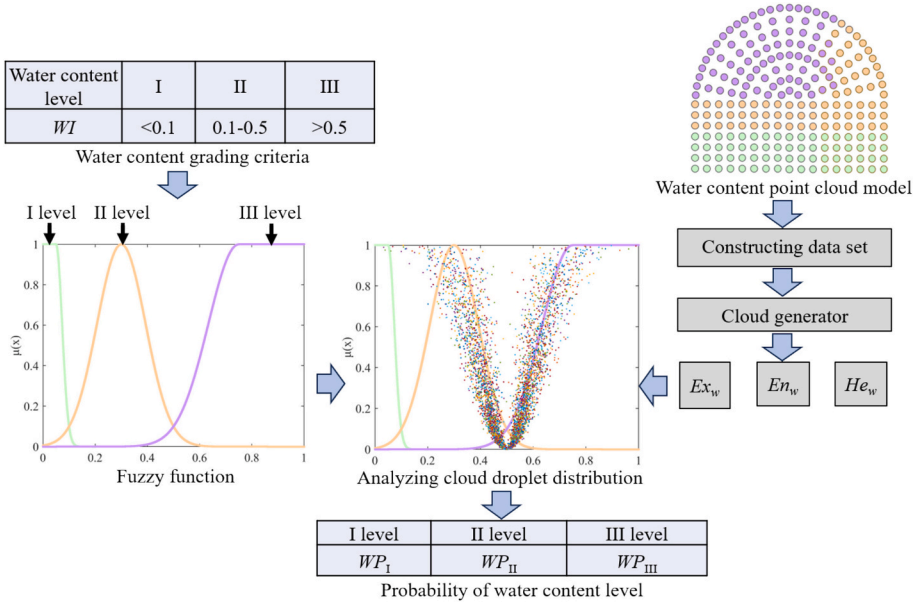


Fig. 7. Intelligent decision-making process for surrounding rock water content level.

$$f_{w,\mathrm{II}} = e^{\left(\frac{x-\frac{0.5+0.1}{2}}{\frac{0.5-0.1}{2}}\right)^2 \ln(0.1)} \tag{8}$$

$$f_{w,\mathrm{III}} = \begin{cases} e^{\left(\frac{x-\frac{1+0.5}{2}}{\frac{1-0.5}{2}}\right)^2 \ln(0.1)} & x < 0.75 \\ 1 & x \geq 0.75 \end{cases} \tag{9}$$

The cloud characteristic values of the water content point cloud model for each meter ahead of the tunnel face are calculated. Using these characteristic values, cloud drops are simulated, and their distribution across different water content levels is analyzed to determine the water content classification for each meter. The workflow is shown in Fig. 7.

Next, the cloud characteristic values for the lithology point cloud model and integrity point cloud model for each meter ahead of the tunnel face are calculated. The characteristic values of the lithology and integrity point cloud models are then combined to calculate the cloud characteristic values of the basic quality of the rock mass using the Basic Quality (BQ) formula. The formula for calculating BQ is as follows:

$$BQ = 100 + 3Rc + 250Kv \tag{10}$$

where, Rc represents the uniaxial compressive strength of the rock, and Kv is the integrity index of the surrounding rock. The cloud characteristic values for BQ are calculated based on Eq. (10), and the relevant calculation formulas are as follows:

$$\begin{cases} Ex_{BQ} = 100 + 3Ex_{Rc} + 250Ex_{Kv} \\ En_{BQ} = 3En_{Rc} + 250En_{Kv} \\ He_{BQ} = 3He_{Rc} + 250He_{Kv} \end{cases} \tag{11}$$

The cloud characteristic values of [BQ] are obtained by modifying BQ based on groundwater conditions, with the groundwater correction factor WK referenced in Table 1. The calculation formula for the cloud characteristic values of [BQ] is as follows:

$$\begin{cases} Ex_{[BQ]} = Ex_{BQ} - 100WK \\ En_{[BQ]} = En_{BQ} \\ He_{[BQ]} = He_{BQ} \end{cases} \tag{12}$$

Similarly, by applying the classification intervals of [BQ] to Eqs. (5) and (6), a normal distribution membership function for rock mass grading is constructed, as expressed below:

$$f_{[BQ],\mathrm{V}} = \begin{cases} 1 & x < 125 \\ e^{\left(\frac{x-\frac{250+0}{2}}{\frac{250-0}{2}}\right)^2 \ln(0.1)} & x \geq 125 \end{cases} \tag{13}$$

$$f_{[BQ],\mathrm{IV}} = e^{\left(\frac{x-\frac{350+250}{2}}{\frac{350-250}{2}}\right)^2 \ln(0.1)} \tag{14}$$

$$f_{[BQ],\mathrm{III}} = e^{\left(\frac{x-\frac{450+350}{2}}{\frac{450-350}{2}}\right)^2 \ln(0.1)} \tag{15}$$

$$f_{[BQ],\mathrm{II}} = e^{\left(\frac{x-\frac{450+350}{2}}{\frac{450-350}{2}}\right)^2 \ln(0.1)} \tag{16}$$

$$f_{[BQ],\mathrm{I}} = \begin{cases} e^{\left(\frac{x-\frac{800+550}{2}}{\frac{800-550}{2}}\right)^2 \ln(0.1)} & x < 625 \\ 1 & x \geq 625 \end{cases} \tag{17}$$

Based on the cloud characteristic values of [BQ] a large number of

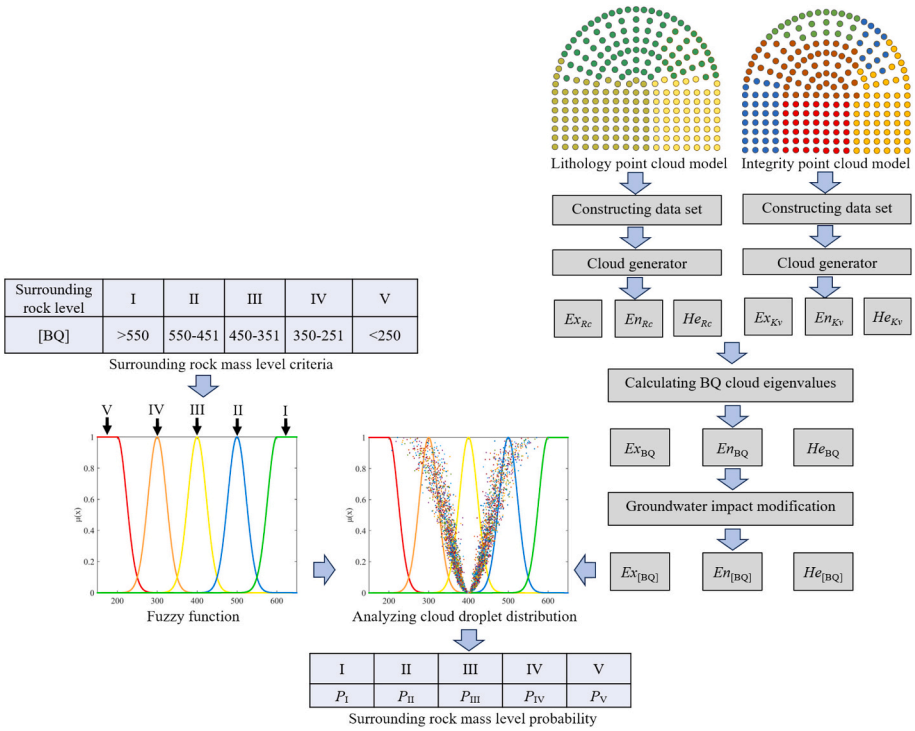


Fig. 8. Intelligent decision-making process for surrounding rock classification.

cloud drops are simulated. The probability of these cloud drops falling into different surrounding rock classification regions is analyzed, and the classification of the surrounding rock is determined according to the principle of maximum probability. The overall decision-making process for surrounding rock classification is illustrated in Fig. 8, and the algorithm workflow is presented in Algorithm 1.

Algorithm 1
Intelligent decision-making for surrounding rock classification using fuzzy cloud probability model.

Input: Water content point cloud model
Lithology point cloud model
Integrity point cloud model
Output: Surrounding rock grade probability: $\{P_{\mathrm{I}}, P_{\mathrm{II}}, P_{\mathrm{III}}, P_{\mathrm{IV}}, P_{\mathrm{V}}\}$
Surrounding rock grade decision

Procedure:
Constructing a water-richness dataset
Generating water-rich cloud eigenvalues (Ex_w, En_w, He_w)
Set $N_{w,i}$=0 (i=I, II, III)
for n = 1 to N **do**
  $Enn_{w,n}$=NORMRND(En_w, He_w)
  $x_{w,n}$=NORMRND(Ex_w, abs($Enn_{w,n}$))
  $z_{w,n}=1\text{-}e^{-\frac{(x_{w,n}-Ex_w)^2}{2Enn_w^2}}$
  if $z_{w,n}<f_{w,\mathrm{I}}$ **then**
    $N_{w,\mathrm{I}}+=1$
  elseif $z_{w,n}<f_{w,\mathrm{II}}$
    $N_{w,\mathrm{II}}+=1$
  elseif $z_{w,n}<f_{w,\mathrm{III}}$
    $N_{w,\mathrm{III}}+=1$
  end if
end for
$WP_i=\frac{N_{w,i}}{\sum_{i=1}^{3}N_{w,i}}$
Decision-making water-richness level is based on Max(WP)
Building integrity dataset and lithologic strength dataset
Generating integrity cloud eigenvalues (Ex_{Kv}, En_{Kv}, He_{Kv})
Generating lithological strength cloud eigenvalues (Ex_{Rc}, En_{Rc}, He_{Rc})
Computing BQ cloud eigenvalues (Ex_{BQ}, En_{BQ}, He_{BQ})
Computing groundwater correction factors (WK)
Computing [BQ] cloud eigenvalues ($Ex_{[\mathrm{BQ}]}$, $En_{[\mathrm{BQ}]}$, $He_{[\mathrm{BQ}]}$)
Set $N_{[\mathrm{BQ}],i}$=0 (i=I, II, III, IV, V)
for n = 1 to N **do**
  $Enn_{[\mathrm{BQ}],n}$=NORMRND($En_{[\mathrm{BQ}]}$, $He_{[\mathrm{BQ}]}$)
  $x_{[\mathrm{BQ}],n}$=NORMRND($Ex_{[\mathrm{BQ}]}$, abs($Enn_{[\mathrm{BQ}],n}$))
  $z_{[\mathrm{BQ}],n}=1\text{-}e^{-\frac{(x_{[\mathrm{BQ}],n}-Ex_{[\mathrm{BQ}]})^2}{2Enn_{[\mathrm{BQ}]}^2}}$
  if $z_{[\mathrm{BQ}],n}<f_{[\mathrm{BQ}],\mathrm{I}}$ **then**
    $N_{[\mathrm{BQ}],\mathrm{I}}+=1$
  elseif $z_{[\mathrm{BQ}],n}<f_{[\mathrm{BQ}],\mathrm{II}}$
    $N_{[\mathrm{BQ}],\mathrm{II}}+=1$
  elseif $z_{[\mathrm{BQ}],n}<f_{[\mathrm{BQ}],\mathrm{III}}$
    $N_{[\mathrm{BQ}],\mathrm{III}}+=1$
  elseif $z_{[\mathrm{BQ}],n}<f_{[\mathrm{BQ}],\mathrm{IV}}$
    $N_{[\mathrm{BQ}],\mathrm{IV}}+=1$
  elseif $z_{[\mathrm{BQ}],n}<f_{[\mathrm{BQ}],\mathrm{V}}$
    $N_{[\mathrm{BQ}],\mathrm{V}}+=1$
  end if
end for
$P_i=\frac{N_{[BQ],i}}{\sum_{i=1}^{5}N_{[BQ],i}}$
Decision-making surrounding rock grade is based on Max(P)

By utilizing the method proposed in this paper to make classification decisions for each meter of surrounding rock within a 30-m range ahead of the tunnel face, both the safety and efficiency of tunnel construction can be significantly improved. The classification decisions help the construction team identify potential risk areas in advance and implement targeted reinforcement measures, effectively preventing sudden geological hazards. Moreover, these decisions aid in the proper scheduling of construction progress and support methods, optimizing resource allocation and minimizing unnecessary waste. With these decisions in place, the construction team can ensure safety while improving work efficiency, ensuring that the project progresses on time and minimizing delays. More importantly, by acquiring real-time rock data and making dynamic adjustments, the construction strategy can be flexibly adapted to geological changes, further enhancing the overall effectiveness of tunnel construction. For a detailed understanding of how rock classification decisions can improve tunnel construction safety and efficiency, refer to our team's research on dynamic support design for tunnels [61].

4. Application case

Section 4 presents the engineering application of the digital twin method for advanced geological conditions in tunnel construction proposed in this paper. This section first introduces the engineering background of the selected case study, followed by a detailed explanation of the entire process of applying the proposed method to this project. Finally, with the help of the iS3 platform, the method is widely applied to five rock tunnels to verify its universality and reliability.

4.1. Engineering background

The Yanjiashai Tunnel is located along the Lei Bo Kaha Luo to Yongshan connection line in Yunnan Province, China, and is considered one of the critical control projects along the entire route. The tunnel is designed with separate passages, with a maximum depth of approximately 933 m. The geological conditions at the tunnel site are highly complex, featuring weak surrounding rocks such as shale and argillaceous limestone. The rock mass is significantly fractured, with well-developed joints and fissures. The tunnel area is prone to various adverse geological phenomena, including landslides, karst development, and high in-situ stress. The hydrogeological conditions are also intricate, with groundwater mainly consisting of bedrock fissure water and karst water, both of which are abundant. The maximum predicted water inflow in the main tunnel is estimated to be 25.37×10^3 m^3/d. The right line extends 7.56×10^3 m, and the left line extends 7.55×10^3 m. High in-situ stress increases the risk of significant deformation in soft rock. These complex geological conditions present substantial challenges to tunnel design and construction, necessitating thorough engineering geological research to develop a scientifically sound construction plan that ensures safety and smooth project progress. The topographic map of the Yongshan area is shown in Fig. 9. Additionally, field images of the experimental tunnel face and the implementation of predictive geological investigations are provided in Fig. 10.

4.2. Application of databases and virtual models

In the engineering case of the Yanjiashai Tunnel, the excavation span of the experimental tunnel face is 9 m, and the excavation height is 10 m. A geological database is constructed based on the collected multi-source geological information, as shown in Fig. 11. Using this geological database, detailed digital twin modeling is performed for the water content, lithology, and integrity of the surrounding rock ahead of the tunnel face.

Based on the TEM-acquired image data, a digital twin model of the water content of the surrounding rock ahead of the tunnel face is constructed. Fig. 12 illustrates the complete digital twin model of the surrounding rock's water content and cross-sectional views for each meter. The model, represented as a three-dimensional point cloud, precisely captures the water content conditions at various locations, thereby aiding engineering professionals in identifying high-risk areas for sudden water inrush. To further improve the model's precision, cross-sectional analysis is performed on the point cloud model for each meter ahead of the tunnel face. These sectional views clearly indicate that the first 10 m ahead of the tunnel face are characterized by a high water content, with a significant portion of the area being classified as Level III water-richness rock, suggesting a high potential risk of sudden water inflow. In the 11 to 22 m range, the water content decreases,

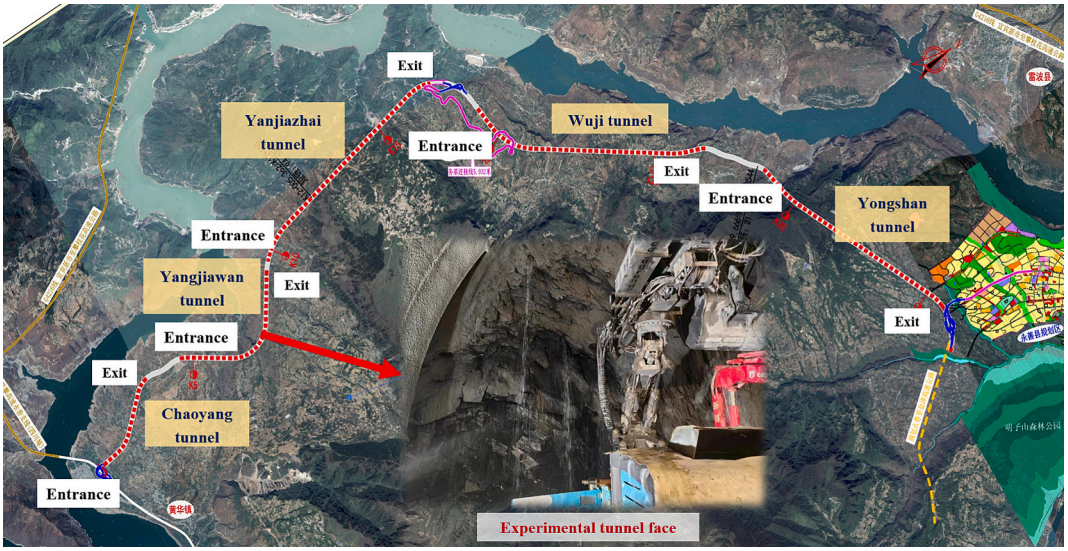


Fig. 9. Topographic map of Yongshan area.

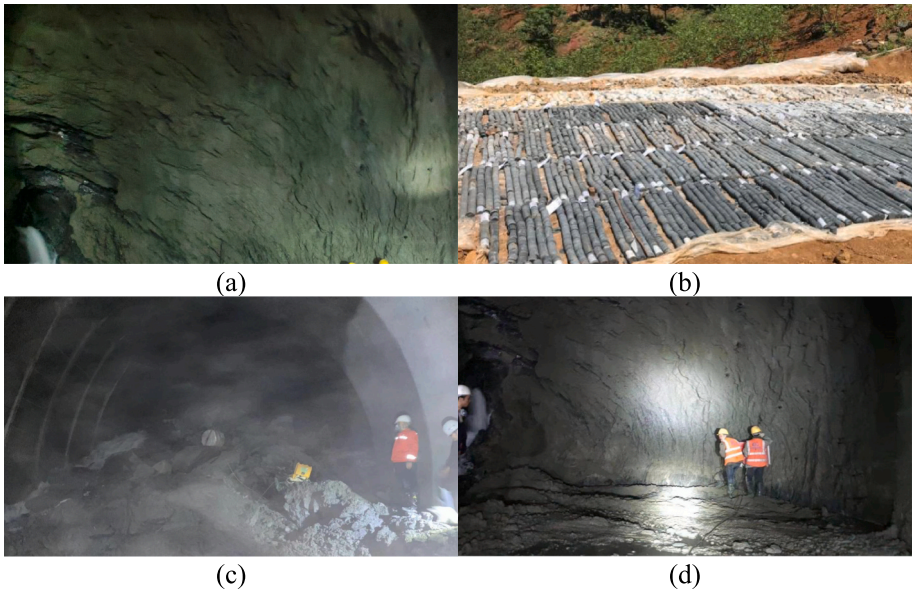


Fig. 10. Actual engineering site; (a) Experimental tunnel face; (b) Drilling core; (c) TEM implementation; (d) GPR implementation.

resulting in a Level II water-richness region. In the 22 to 30 m range, the water content continues to decrease, with Level I water-richness regions gradually becoming dominant.

Data from five tunnels in the Yongshan area, including borehole data, excavation exposure data, and tunnel geological profile maps, are collected and processed to construct a lithology database, as shown in Fig. 11(a). Based on this lithology database, geological models of the mountain along the Yanjiashai Tunnel alignment and along the tunnel axis are constructed using KNN interpolation. Using the spatial coordinates of the experimental tunnel face in this study, a lithology model of the surrounding rock ahead of the tunnel face is constructed, as depicted in Fig. 13. The model indicates that the lithology of the surrounding rock ahead of the tunnel face is shale. According to the local field conditions, the uniaxial compressive strength of the shale is 30 MPa, which is used for subsequent intelligent decisions regarding rock classification. GPR images collected on-site are processed to extract integrity information of the surrounding rock ahead of the tunnel face, as shown in Fig. 11(c). Using this extracted integrity information, a model of the rock integrity ahead of the tunnel face is constructed. It can be observed that the rock integrity is generally good for the first 1–19 m, whereas the rock is more fractured between 20 and 29 m, particularly at the 21 and 22-m sections, where there are significantly large areas of highly fractured rock. Special attention should be paid to preventing collapse incidents during construction in these fractured areas, as illustrated in Fig. 14.

4.3. *Application of surrounding rock classification decision-making*

In determining the surrounding rock classification for each meter ahead of the tunnel face based on the three constructed digital twin models, a fuzzy cloud probability model is first employed to quantify the uncertainty of the surrounding rock's water content and to decide its water content classification. Fig. 15 shows the cloud maps for each meter within the 30-m range. As distance increases, the center of the cloud model gradually shifts to the left, indicating a gradual decline in the overall water content of the surrounding rock. Between 2 and 12 m,

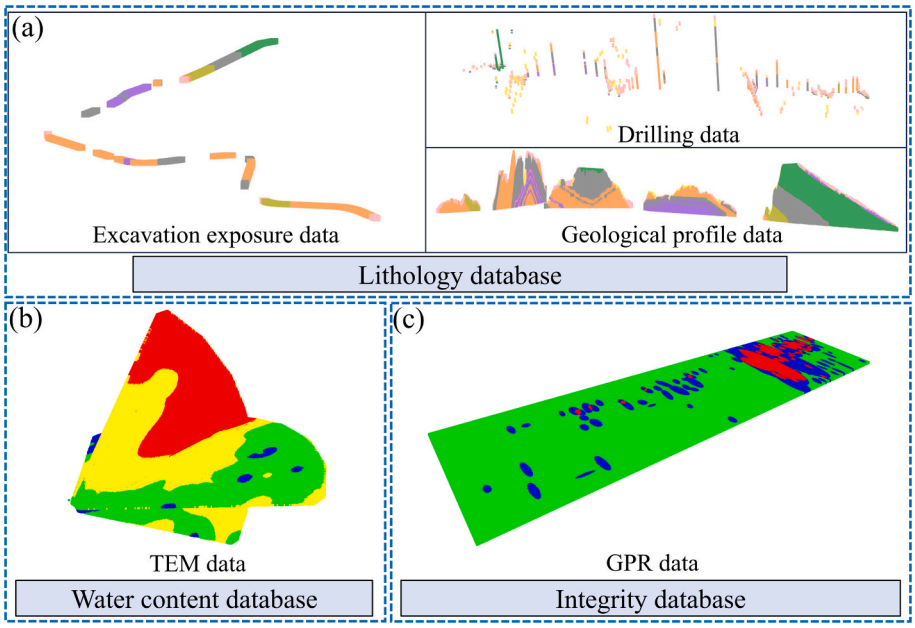


Fig. 11. Constructed database.

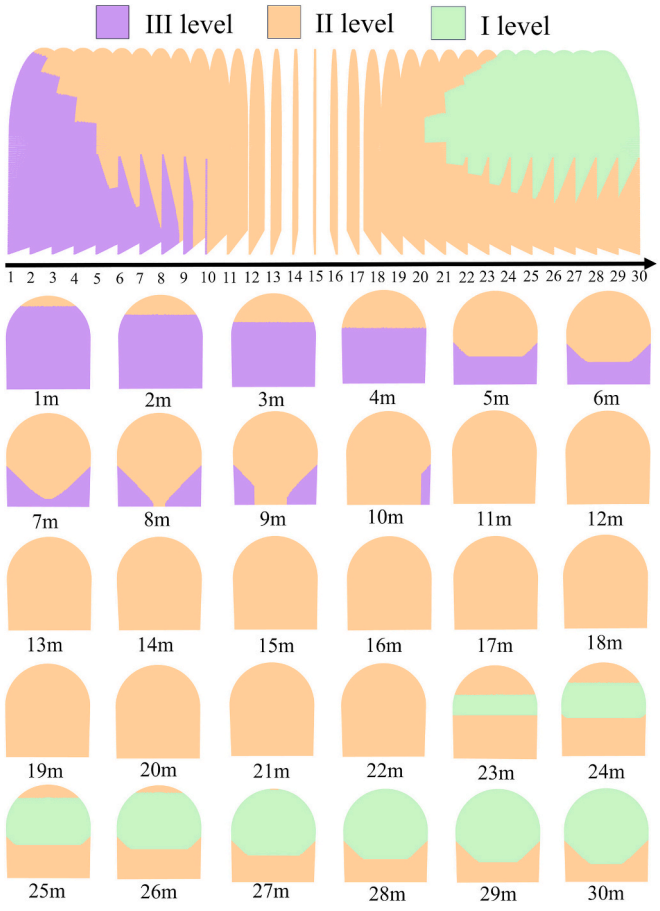


Fig. 12. Digital twin model of surrounding rock water content.

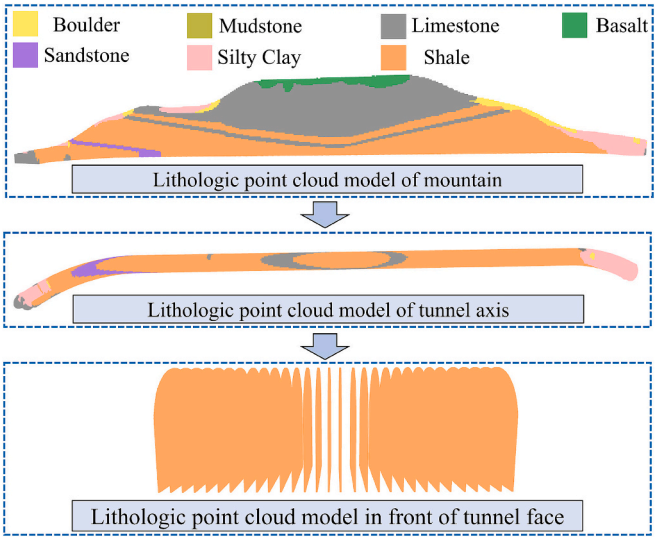


Fig. 13. Digital twin model of surrounding rock lithology.

the cloud drops are relatively dispersed, reflecting significant uncertainty in the water content of the surrounding rock within this range. In contrast, in the 18–22 m range, the cloud drops are more concentrated, suggesting relatively low uncertainty in the rock's water content. Fig. 16 illustrates how the probability of different water content levels, as quantified by the fuzzy cloud probability model, changes with increasing distance. Table 2 presents the quantified data for each meter of the water content digital twin model within the 1–30 m range ahead of the tunnel face, as determined by the fuzzy cloud probability model. The probability of WP_{III} decreases progressively with increasing distance and drops to zero at 16 m. Meanwhile, WP_{I} begins to increase from 21 m onward, reaching 0.81 at 30 m. Based on these quantified probability data, the water content classification is determined for each meter. The

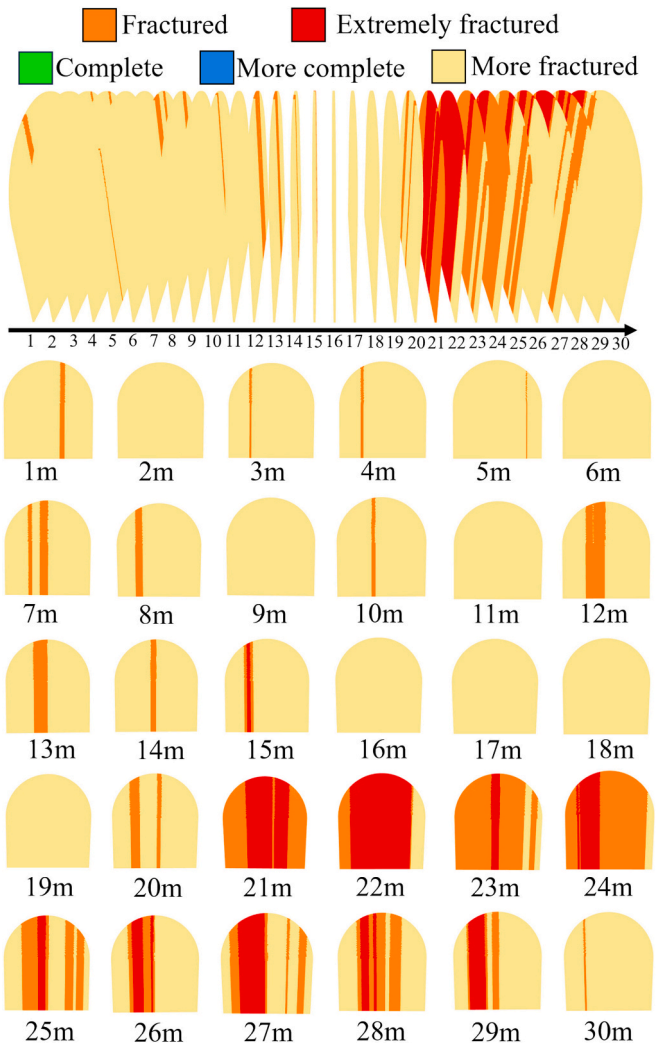


Fig. 14. Digital twin model of surrounding rock integrity.

range from 1 to 5 m is classified as Level III water content, the range from 6 to 22 m as Level II, and the range from 23 to 30 m as Level I water content.

The cloud characteristic values of the surrounding rock integrity are calculated for each meter ahead of the tunnel face. Since the lithology within the 30-m range ahead of the tunnel face is all shale, the BQ cloud characteristic value for each meter is computed based on its uniaxial compressive strength and the integrity cloud characteristic value. Using the decided water content classification and BQ cloud characteristic values, the groundwater correction factor is calculated through linear interpolation as outlined in Table 1. The data relevant to this calculation process is presented in Table 3. Using the groundwater correction factor and BQ cloud characteristic values, the corrected [BQ] cloud characteristic value is computed, taking into account the influence of groundwater on the original BQ cloud characteristic values. Based on the computed [BQ] cloud characteristic values, cloud drops are simulated using the cloud generator. Fig. 17 shows the [BQ] cloud map for each meter ahead of the tunnel face. The cloud drops are more dispersed between 25 and 28 m, indicating higher uncertainty in the surrounding rock in this range. In contrast, the cloud drops are more concentrated between 22 and 24 m, suggesting that the properties of the surrounding rock in this section are relatively consistent.

The distribution of cloud drops across different surrounding rock classification regions is analyzed to quantify the probability of each surrounding rock classification, as shown in Fig. 18. The surrounding rock classification is ultimately determined based on the highest probability, and the relevant data are presented in Table 4. The results indicate that the 1–5 m range, the 20–28 m range, and at 12 m are classified as Level V surrounding rock. This outcome is largely due to the high water content in the 1–5 m range and the high degree of rock fragmentation between 20 and 28 m. To further validate the accuracy of the intelligent decision-making, the rock conditions ahead of the tunnel face at 3 m and 24 m are recorded as construction continues, as shown in Fig. 19. At 3 m, the surrounding rock experienced a significant sudden water inrush, and at 24 m, the rock was highly fractured. These observations are consistent with the phenomena reflected in the digital twin model, and the surrounding rock classifications calculated on-site match the classifications determined by the intelligent decision-making process.

4.4. Universality and reliability verification

To verify the universality of the tunnel construction data-model-decision closed-loop advanced geological digital twin method proposed in this paper, this study utilizes an intelligent tunnel construction platform developed based on digital twin technology for complex mountainous conditions, using the iS3 platform, as shown in Fig. 20. The iS3 platform, constructed by the team led by Academician Zhu Hehua, is a decision service system that integrates infrastructure lifecycle data collection, processing, expression, and analysis. The algorithm introduced in this paper serves as the core service of this platform. Through this platform, the algorithm can not only be widely applied across multiple tunnel projects but also enable real-time acquisition of relevant data from various tunnels and provide rapid feedback.

Specifically, the iS3 platform can upload real-time advanced geological forecast data and other relevant information directly from the construction site. This data encompasses geological details, monitoring data, and tunnel construction progress, as illustrated in Fig. 21(a). By employing the algorithm proposed in this paper, rapid decision-making regarding the geological conditions ahead of the tunnel face is enabled, thereby providing the construction team with reliable geological forecast information, as shown in Fig. 21(b). The platform has already been successfully applied in several rock tunnel projects. With the broader implementation of the iS3 platform, the geological digital twin framework introduced in this paper has been deployed in numerous mountainous rock tunnel projects, validating its universality and effectiveness in complex geological environments.

To further validate the reliability of the proposed framework, a validation scheme is designed to compare the rock mass classification decision results obtained through the geological digital twin method before tunnel excavation with the actual rock mass classifications exposed after excavation, thereby verifying the accuracy and effectiveness of the advanced rock mass condition decision-making. First, to conduct large-scale testing and validation, a total of 539 exposed rock mass classification data points from five rock tunnel projects recorded by the platform are collected. These data come from different locations and depths of the tunnel face, covering a variety of complex rock types and geological conditions. The exposed rock mass classification data are

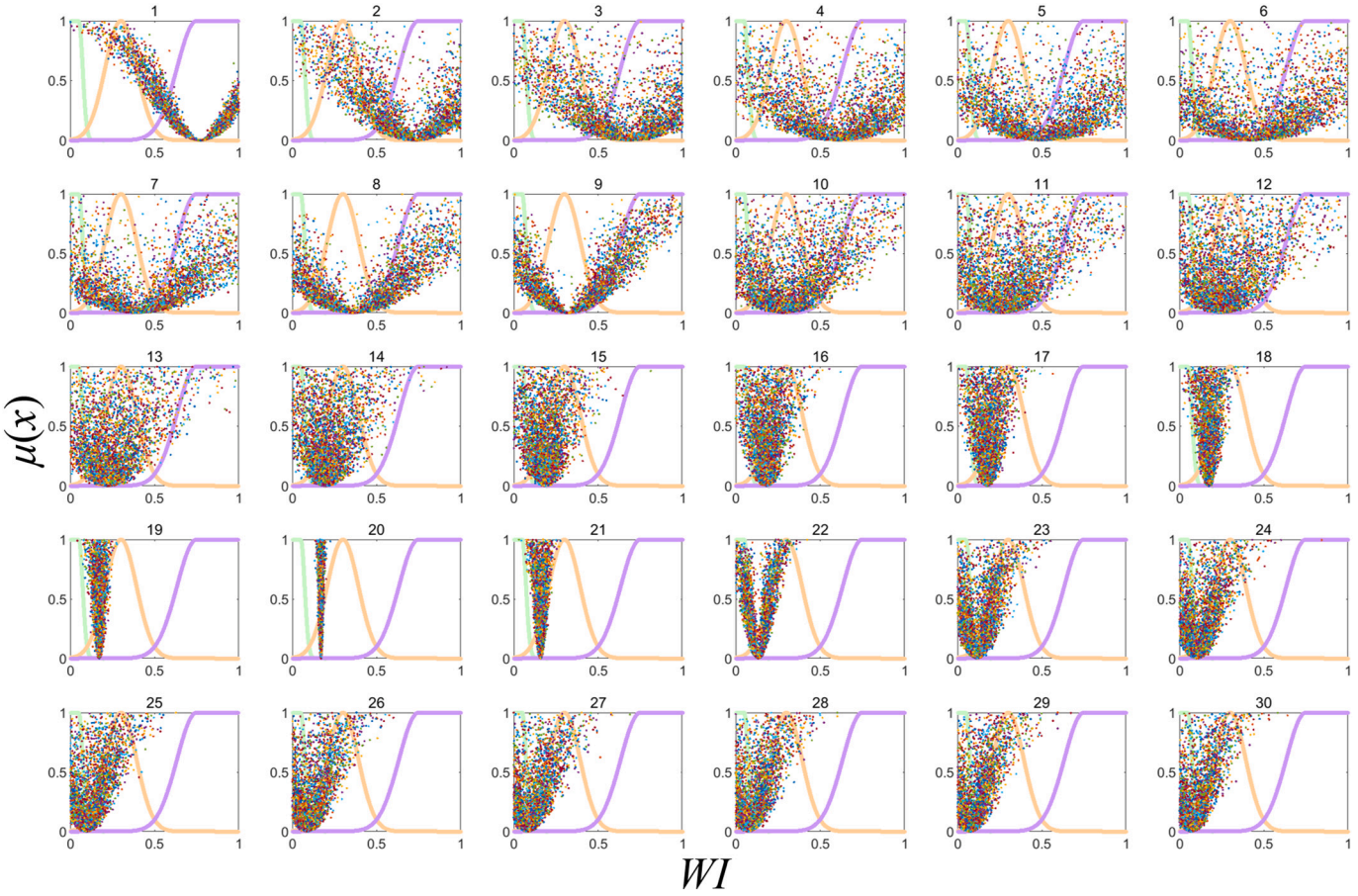


Fig. 15. Water content cloud map for each meter from 1 to 30 m ahead of tunnel face.

carefully recorded and stored on the platform as representations of the actual geological conditions after excavation, as shown in Fig. 21(c). Next, based on these actual exposed data, the results are compared with the corresponding advanced rock mass classification decision results. These decision results are obtained through the geological digital twin method before the tunnel excavation. By comparing the actual rock mass classification at each sampling point with the advanced decision classification, the overall decision accuracy is calculated. The accuracy is then used to assess the reliability of the method proposed in this paper.

This method is validated using five rock tunnel engineering projects: Yanjiazhai Tunnel, Wuji Tunnel, Yongshan Tunnel, Chaoyang Tunnel, and Yangjiawan Tunnel. The validation results indicate that the Yanjiazhai Tunnel includes 122 data points with an accuracy of 93.44 %, Wuji Tunnel has 83 data points with an accuracy of 92.77 %, Yongshan Tunnel consists of 125 data points with an accuracy of 93.60 %, Chaoyang Tunnel has 115 data points with an accuracy of 90.43 %, and Yangjiawan Tunnel contains 94 data points with an accuracy of 94.68 %. Collectively, the five tunnels account for a total of 539 data points, achieving an overall accuracy of 92.95 %, as summarized in Table 5. The corresponding confusion matrix is shown in Fig. 22. These results demonstrate that the digital twin framework proposed in this paper achieves satisfactory performance across all five rock tunnel projects, confirming its universality and reliability in practical engineering applications.

5. Discussion

The discussion section mainly explores the applicability, computational demands, shortcomings, and future research suggestions of the proposed advanced geological digital twin method. It is hoped that more scholars will build upon the method presented in this paper and, based on the suggestions provided in this section, continue to explore related research fields.

5.1. Applicability

The advanced geological digital twin method for tunnel construction proposed in this paper has been successfully applied in several mountain tunnel projects through an intelligent tunnel construction platform, where the geological environment is primarily rock mass. The developed framework can quickly analyze the geological conditions ahead of the tunnel face and make classification decisions for the surrounding rock at each meter. However, the applicability of the method in other types of geological environments requires further discussion. For instance, during the construction of underwater tunnels, the influence of the surrounding seawater medium raises concerns about the applicability of the TEM method. Additionally, in the excavation of subway tunnels in soft soil environments, the adaptability of the existing method to soil layers needs to be validated. In these specific cases, it may be necessary to

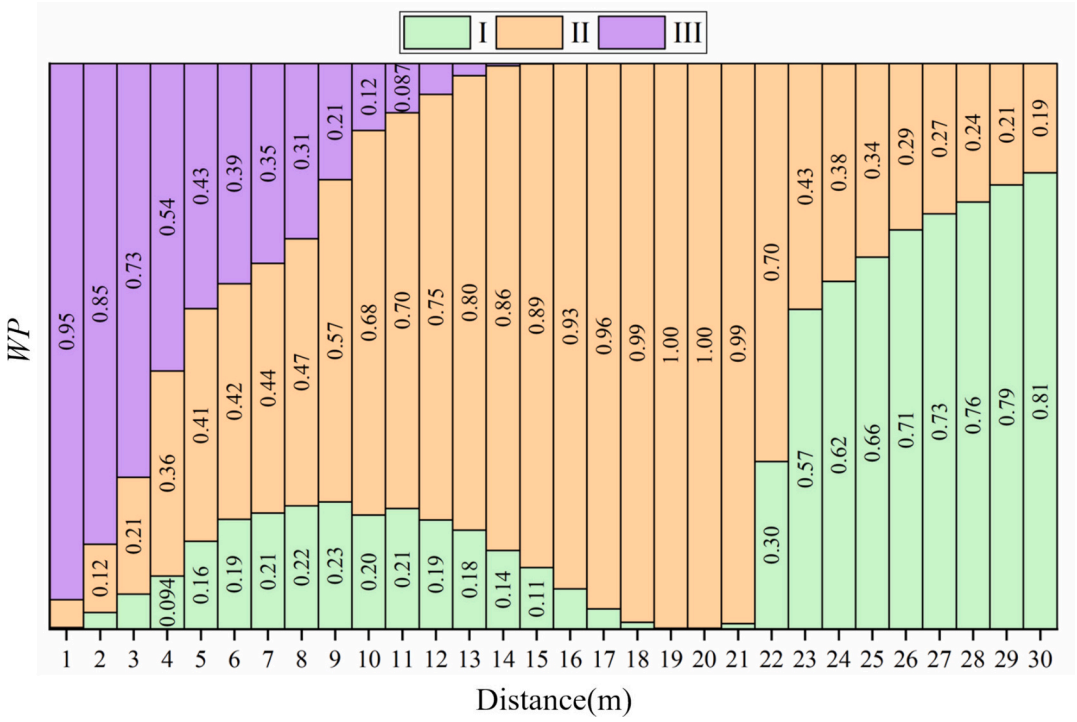


Fig. 16. Probability distribution of surrounding rock water content classification.

Table 2
Quantified data of surrounding rock water content.

D(m)	Ex_w	En_w	He_w	WP_{I}	WP_{II}	WP_{III}	Max(WP)
1	0.7696	0.2532	0.0387	0.0031	0.0494	0.9475	III
2	0.7295	0.3072	0.1173	0.0295	0.1211	0.8494	III
3	0.6782	0.3553	0.1662	0.0623	0.2066	0.7311	III
4	0.6009	0.3775	0.1829	0.0943	0.3624	0.5433	III
5	0.5032	0.3971	0.1913	0.1553	0.4118	0.4329	III
6	0.4323	0.4221	0.2087	0.1944	0.4163	0.3893	II
7	0.3957	0.3911	0.1680	0.2054	0.4414	0.3531	II
8	0.3633	0.3493	0.1130	0.2183	0.4721	0.3096	II
9	0.3222	0.2792	0.0761	0.2251	0.5696	0.2053	II
10	0.2847	0.2221	0.1394	0.2022	0.6796	0.1181	II
11	0.2598	0.1907	0.1501	0.2134	0.6999	0.0866	II
12	0.2473	0.1552	0.1486	0.1936	0.7522	0.0543	II
13	0.2256	0.1082	0.1312	0.1756	0.8031	0.0213	II
14	0.1989	0.0644	0.1059	0.1394	0.8567	0.0039	II
15	0.1863	0.0382	0.0781	0.1089	0.8908	0.0002	II
16	0.1814	0.0271	0.0614	0.0714	0.9286	0.0000	II
17	0.1777	0.0190	0.0480	0.0362	0.9638	0.0000	II
18	0.1734	0.0108	0.0319	0.0122	0.9878	0.0000	II
19	0.1712	0.0053	0.0170	0.0020	0.9980	0.0000	II
20	0.1701	0.0015	0.0057	0.0018	0.9982	0.0000	II
21	0.1595	0.0173	0.0107	0.0099	0.9901	0.0000	II
22	0.1339	0.0567	0.0191	0.2968	0.7032	0.0000	II
23	0.1155	0.0804	0.0407	0.5657	0.4343	0.0000	I
24	0.1045	0.0880	0.0487	0.6153	0.3845	0.0002	I
25	0.0960	0.0913	0.0521	0.6578	0.3422	0.0000	I
26	0.0891	0.0928	0.0535	0.7059	0.2941	0.0000	I
27	0.0828	0.0926	0.0533	0.7344	0.2656	0.0000	I
28	0.0777	0.0913	0.0520	0.7555	0.2445	0.0000	I
29	0.0740	0.0908	0.0511	0.7855	0.2145	0.0000	I
30	0.0699	0.0890	0.0489	0.8068	0.1932	0.0000	I

introduce other ahead-of-face geological prediction methods to obtain more comprehensive geological information, and the original method structure should also be adapted and improved according to the specific geological conditions.

5.2. Computational demands and potential latency issues

The geological digital twin method proposed in this paper has shown promising results in practical applications. However, when applied to large-scale or multiple tunnel projects running simultaneously, it faces significant computational demands and potential latency issues. For example, in the Yanjiazhai Tunnel, real-time decision-making involves processing millions of 3D geological point cloud data points, with KNN interpolation requiring approximately 10^6 to 10^7 distance computations per cycle, and the fuzzy cloud probability model generating around 10,000 cloud drops, further increasing computational complexity. On-site tests with standard CPU architecture show that each interpolation and cloud modeling cycle takes between 10 and 60 s. However, when multiple tunnels are operated simultaneously, the computational delay may increase from 2 min to 5 min, affecting the timeliness of real-time decisions.

To address these issues, three technical solutions are proposed: (1) GPU acceleration and parallel computing, where using NVIDIA Tesla GPUs can accelerate KNN interpolation by 10 times and fuzzy cloud model calculations by 20 times; (2) Efficient spatial indexing structures like KD-trees can reduce the complexity of KNN interpolation from O(n) to O(log n), significantly lowering computational load; (3) An edge-cloud collaborative computing architecture, where edge devices rapidly process basic data, and computationally intensive tasks are

Table 3
Surrounding rock integrity, BQ cloud characteristic values, and groundwater correction factor.

D(m)	Ex_{Kv}	En_{Kv}	He_{Kv}	Ex_{BQ}	En_{BQ}	He_{BQ}	*WL*	*WK*
1	0.5418	0.0197	0.0395	325.4444	4.9318	9.8856	III	0.7491
2	0.5452	0.0114	0.0224	326.3083	2.8561	5.6108	III	0.7474
3	0.5408	0.0215	0.0330	325.1985	5.3684	8.2550	III	0.7496
4	0.5442	0.0140	0.0305	326.0410	3.5038	7.6373	III	0.7479
5	0.5431	0.0157	0.0187	325.7631	3.9189	4.6674	III	0.7485
6	0.5117	0.0739	0.0408	317.9240	18.4822	10.2059	II	0.4642
7	0.5064	0.0872	0.0390	316.5947	21.7963	9.7405	II	0.4668
8	0.5339	0.0350	0.0421	323.4848	8.7378	10.5276	II	0.4530
9	0.5373	0.0275	0.0292	324.3233	6.8860	7.2910	II	0.4514
10	0.5406	0.0213	0.0313	325.1617	5.3316	7.8252	II	0.4497
11	0.5077	0.0866	0.0471	316.9183	21.6496	11.7680	II	0.4662
12	0.4715	0.1357	0.0547	307.8745	33.9189	13.6871	II	0.4843
13	0.4947	0.1042	0.0249	313.6795	26.0418	6.2225	II	0.4726
14	0.5129	0.0747	0.0575	318.2371	18.6801	14.3800	II	0.4635
15	0.5302	0.0441	0.0600	322.5541	11.0133	14.9912	II	0.4549
16	0.5500	0.0000	0.0000	327.5000	0.0000	0.0000	II	0.4450
17	0.5500	0.0000	0.0000	327.5000	0.0000	0.0000	II	0.4450
18	0.5494	0.0015	0.0053	327.3504	0.3685	1.3140	II	0.4453
19	0.5108	0.0668	0.0309	317.6902	16.7076	7.7308	II	0.4646
20	0.3170	0.2030	0.1026	269.2375	50.7587	25.6417	II	0.5615
21	0.1589	0.0971	0.0413	229.7276	24.2861	10.3316	II	0.7162
22	0.2022	0.1246	0.0114	240.5562	31.1403	2.8538	II	0.7076
23	0.2356	0.0736	0.0594	248.9024	18.3999	14.8419	I	0.4009
24	0.2907	0.1491	0.0254	262.6724	37.2756	6.3525	I	0.2873
25	0.3845	0.1981	0.1035	286.1148	49.5278	25.8683	I	0.2639
26	0.3683	0.2091	0.1026	282.0853	52.2794	25.6530	I	0.2679
27	0.3412	0.1950	0.0911	275.2976	48.7581	22.7755	I	0.2747
28	0.3872	0.1957	0.0994	286.8006	48.9314	24.8385	I	0.2632
29	0.4695	0.1422	0.0168	307.3845	35.5459	4.2114	I	0.2426
30	0.5283	0.0467	0.0397	322.0628	11.6702	9.9258	I	0.2279

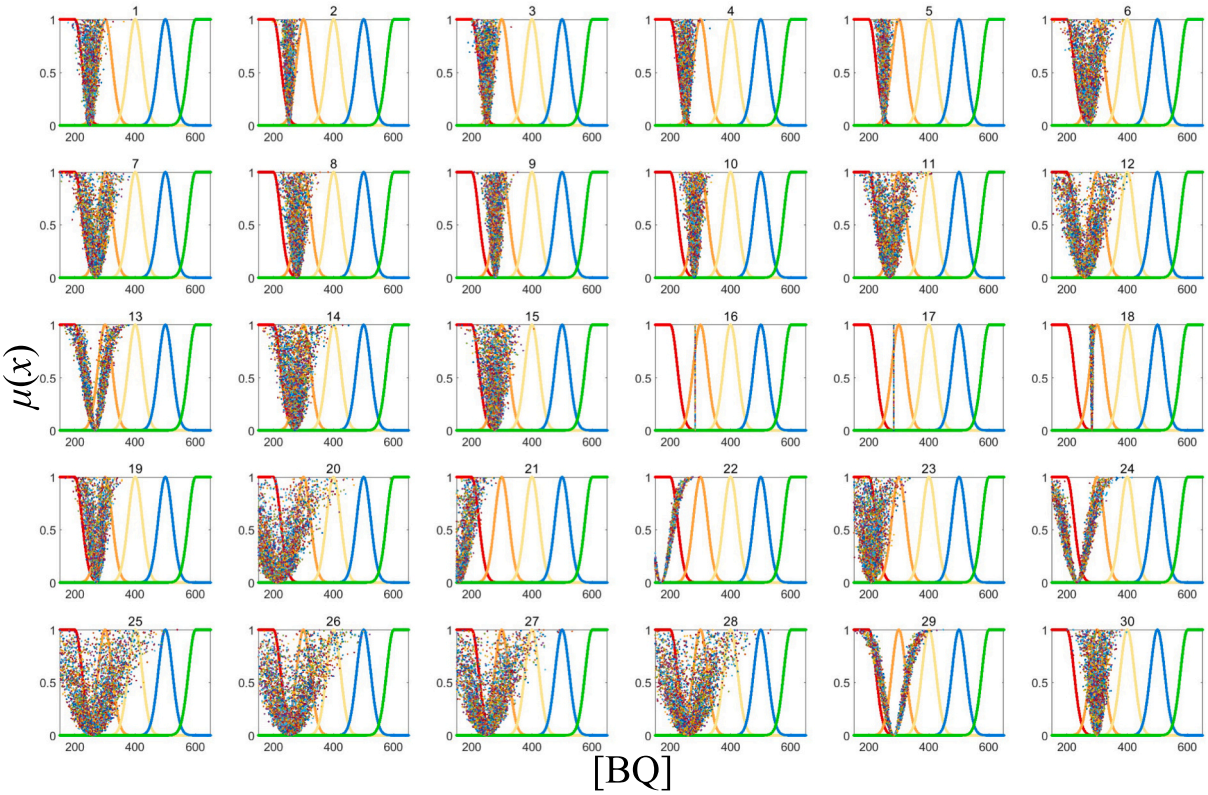


Fig. 17. [BQ] cloud map for each meter from 1 to 30 m ahead of tunnel face.

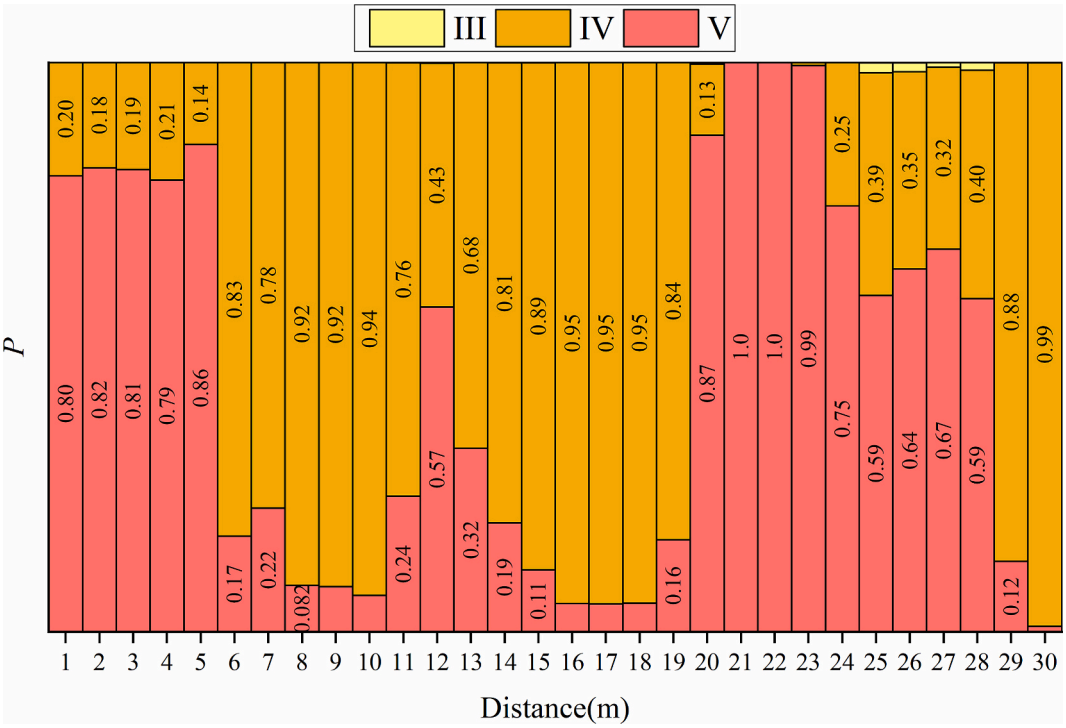


Fig. 18. Probability distribution of surrounding rock classification.

Table 4
[BQ] quantified data.

D(m)	$Ex_{[BQ]}$	$En_{[BQ]}$	$He_{[BQ]}$	P_{I}	P_{II}	P_{III}	P_{IV}	P_{V}	Max(P)
1	250.5333	4.9318	9.8856	0.0000	0.0000	0.0000	0.1988	0.8012	V
2	251.5700	2.8561	5.6108	0.0000	0.0000	0.0000	0.1848	0.8152	V
3	250.2382	5.3684	8.2550	0.0000	0.0000	0.0000	0.1876	0.8124	V
4	251.2492	3.5038	7.6373	0.0000	0.0000	0.0000	0.2062	0.7938	V
5	250.9157	3.9189	4.6674	0.0000	0.0000	0.0000	0.1440	0.8560	V
6	271.5088	18.4822	10.2059	0.0000	0.0000	0.0000	0.8319	0.1681	IV
7	269.9136	21.7963	9.7405	0.0000	0.0000	0.0000	0.7824	0.2176	IV
8	278.1818	8.7378	10.5276	0.0000	0.0000	0.0000	0.9182	0.0818	IV
9	279.1879	6.8860	7.2910	0.0000	0.0000	0.0000	0.9202	0.0798	IV
10	280.1940	5.3316	7.8252	0.0000	0.0000	0.0000	0.9355	0.0645	IV
11	270.3020	21.6496	11.7680	0.0000	0.0000	0.0003	0.7612	0.2385	IV
12	259.4494	33.9189	13.6871	0.0000	0.0000	0.0009	0.4281	0.5710	V
13	266.4154	26.0418	6.2225	0.0000	0.0000	0.0000	0.6772	0.3228	IV
14	271.8845	18.6801	14.3800	0.0000	0.0000	0.0003	0.8081	0.1916	IV
15	277.0649	11.0133	14.9912	0.0000	0.0000	0.0003	0.8904	0.1094	IV
16	283.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.9501	0.0499	IV
17	283.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.9504	0.0496	IV
18	282.8205	0.3685	1.3140	0.0000	0.0000	0.0000	0.9493	0.0507	IV
19	271.2283	16.7076	7.7308	0.0000	0.0000	0.0000	0.8378	0.1622	IV
20	213.0850	50.7587	25.6417	0.0000	0.0000	0.0025	0.1251	0.8724	V
21	158.1059	24.2861	10.3316	0.0000	0.0000	0.0000	0.0000	1.0000	V
22	169.8007	31.1403	2.8538	0.0000	0.0000	0.0000	0.0000	1.0000	V
23	208.8146	18.3999	14.8419	0.0000	0.0000	0.0000	0.0053	0.9947	V
24	233.9396	37.2756	6.3525	0.0000	0.0000	0.0000	0.2515	0.7485	V
25	259.7263	49.5278	25.8683	0.0000	0.0002	0.0175	0.3908	0.5915	V
26	255.2938	52.2794	25.6530	0.0000	0.0005	0.0157	0.3463	0.6376	V
27	247.8273	48.7581	22.7755	0.0000	0.0000	0.0078	0.3198	0.6724	V
28	260.4807	48.9314	24.8385	0.0000	0.0000	0.0134	0.4010	0.5856	V
29	283.1229	35.5459	4.2114	0.0000	0.0000	0.0003	0.8755	0.1242	IV
30	299.2691	11.6702	9.9258	0.0000	0.0000	0.0002	0.9895	0.0103	IV

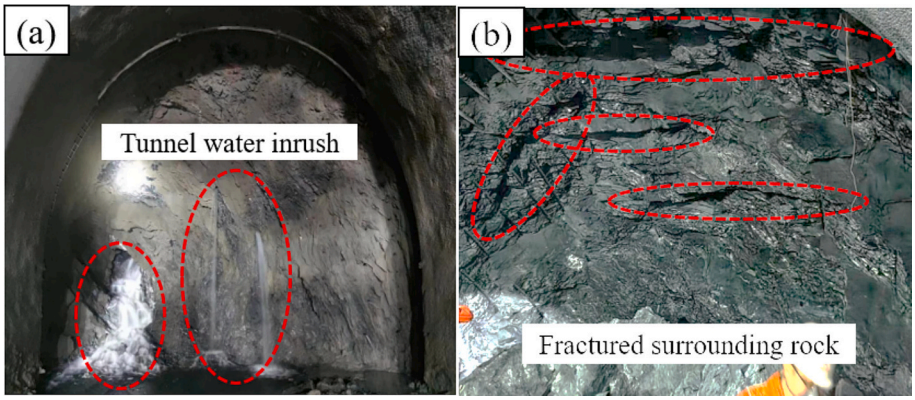
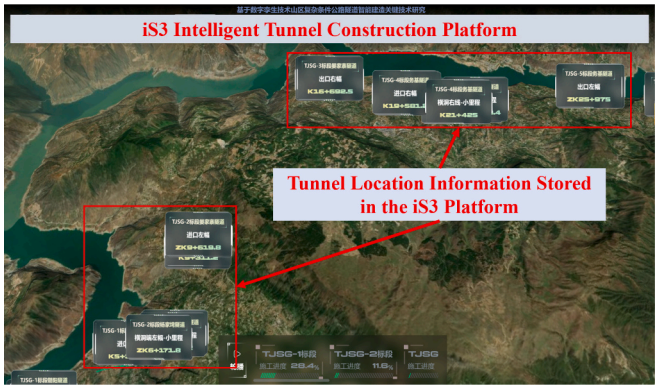
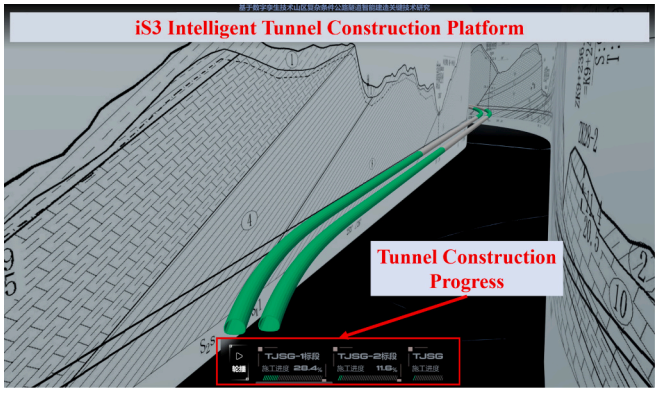


Fig. 19. Surrounding rock condition validation; (a) Surrounding rock at 3 m; (b) Surrounding rock at 24 m.



(a)



(b)

Fig. 20. Intelligent tunnel construction platform based on digital twin technology for complex mountainous conditions.

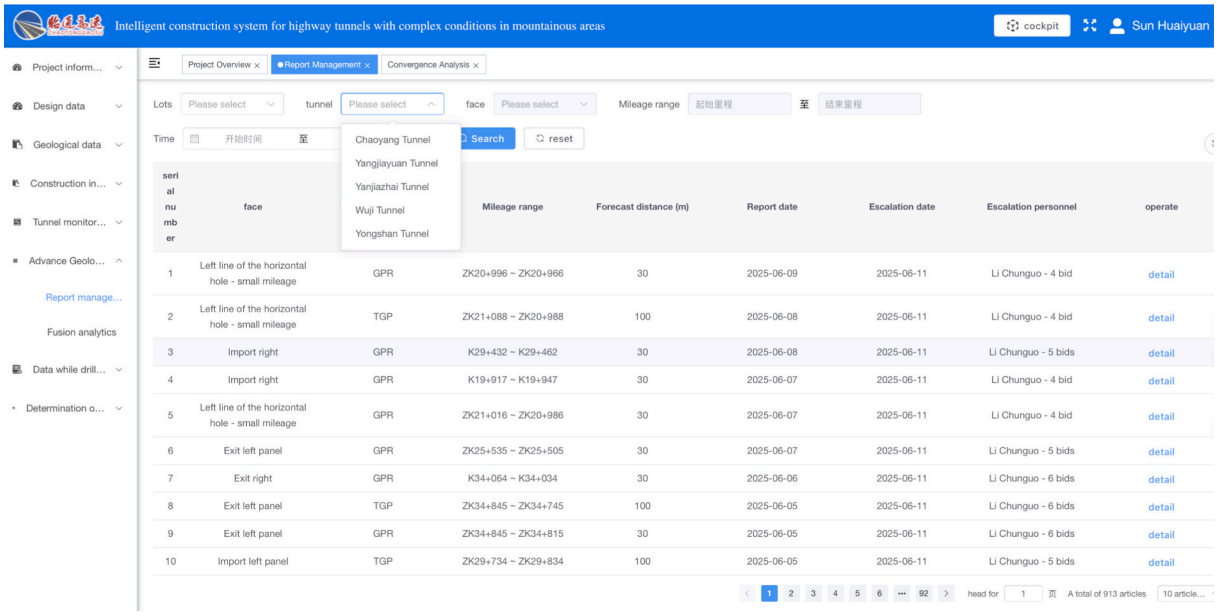
offloaded to cloud-based high-performance platforms, further reducing latency and improving response speed.

5.3. Shortcomings

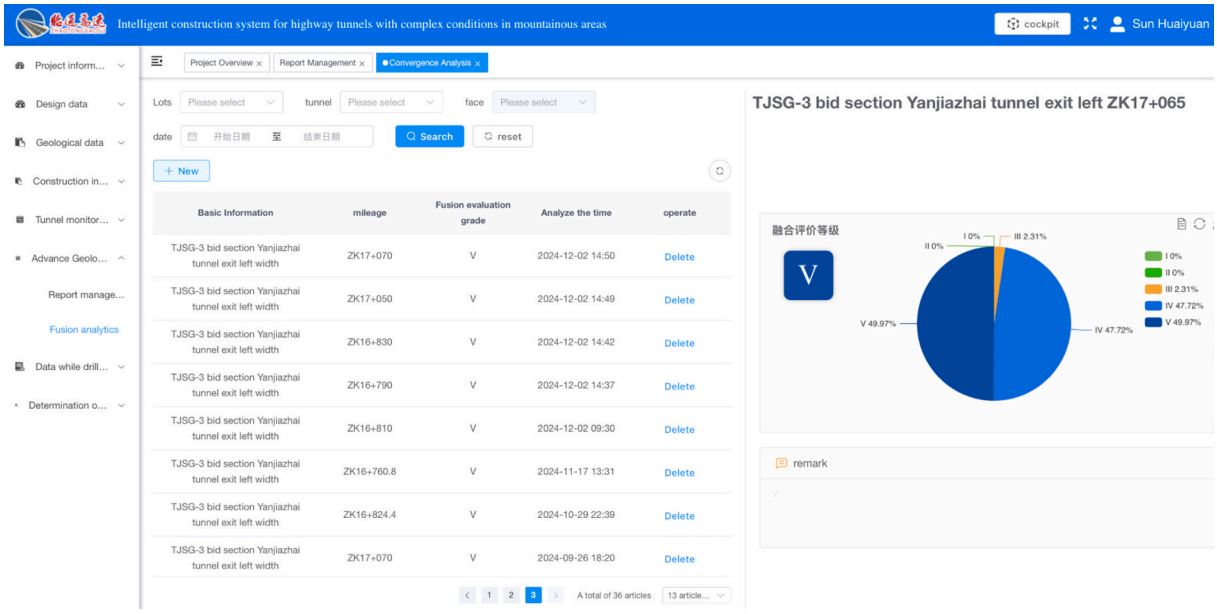
The digital twin method proposed in this paper combines multi-source data fusion, point cloud visualization, and fuzzy cloud probability models, addressing the issues of incomplete information and difficulty in quantitative analysis in traditional geological forecasting methods. It can be used to effectively assess the geological conditions in tunnel construction. However, the model still has some limitations: (1) Improvement in decision-making factors for surrounding rock classification: The digital twin models consider three factors (water content, lithology, and integrity) to classify the surrounding rock. However, more comprehensive intelligent decision-making should take into account additional factors that influence the surrounding rock conditions, such as the effect of structural plane orientations and initial stress conditions. (2) Limitations of GPR image data: The waveform images from GPR can only reflect the fragmentation of the surrounding rock in the horizontal cross-section ahead of the tunnel face. This results in the database containing only the rock integrity information from a single cross-sectional plane, and the digital twin model lacks information on the integrity variations in the longitudinal direction. (3) Uneven database sizes: Currently, the geological databases constructed exhibit significant differences in scale. For example, the lithology database includes borehole data, geological profile maps, and geological information exposed during excavation, resulting in a relatively large data volume. In contrast, the integrity database only contains surrounding rock integrity information obtained via ground-penetrating radar within a limited range of 30 m ahead of the tunnel face, making its data volume relatively small. Such unevenness in database size leads to inconsistencies in the effective dimensions of geological digital twin models constructed from each dataset. In this study, the geological digital twin model uniformly covers a 30-m range ahead of the tunnel face, providing sufficient geological reference information to guide the next tunneling cycle and meeting current research requirements. However, achieving balanced database sizes remains an important consideration when constructing multi-dimensional, large-scale geological digital twin models.

6. Conclusions

This paper presented a digital twin method for mapping the geological conditions ahead of the tunnel face in tunnel construction. The method constructs multiple virtual models from various perspectives to virtually represent the geological conditions ahead of the tunnel face and make intelligent decisions for rock classification, thus achieving a closed-loop of data-model-decision. Developed using the iS3 platform, this method has been successfully applied to case study and several other rock tunnel projects. The findings of the paper include: (1) The proposed multi-source data collection method enables the rapid acquisition of surrounding rock data related to water content, lithology,

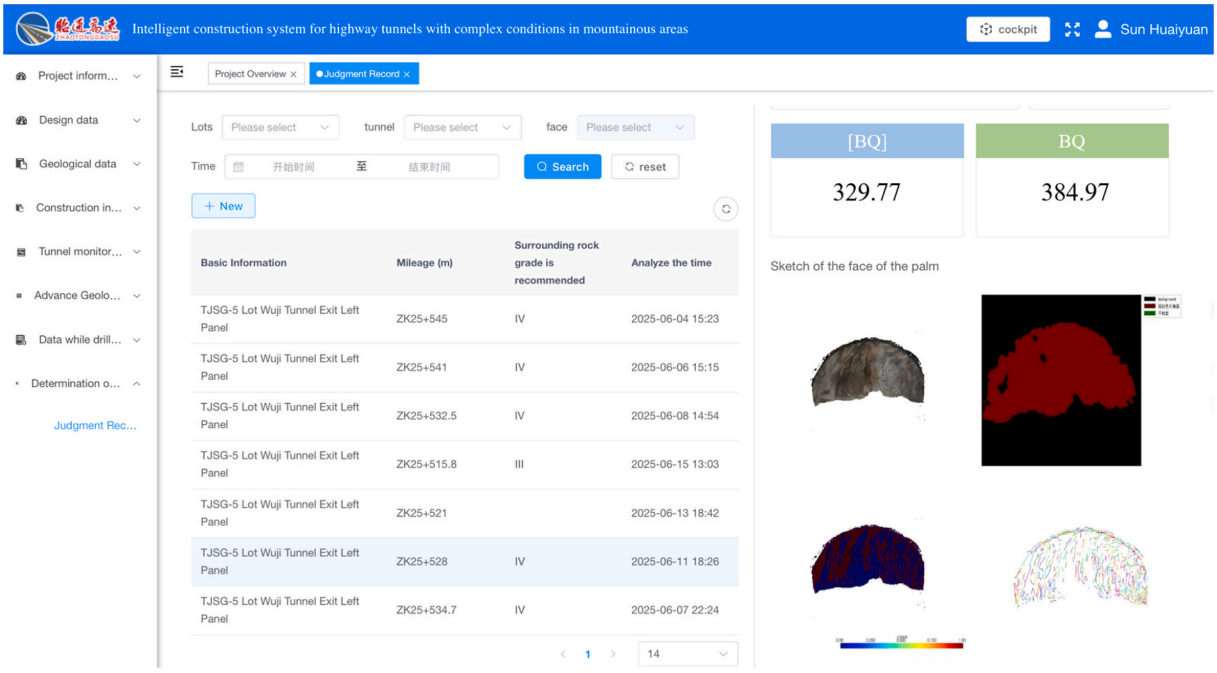


(a)



(b)

Fig. 21. Application of the intelligent construction platform; (a) real-time data from the platform; (b) intelligent decision-making for advanced surrounding rock classification; (c) calculation of surrounding rock classification at excavated tunnel face.



(c)

Fig. 21. (*continued*).

Table 5
Verification of data volume and accuracy for five tunnels.

Tunnel	Yanjiazhai tunnel	Wuji tunnel	Yongshan tunnel	Chaoyang tunnel	Yangjiawan tunnel	The total sum of five tunnels
Verify data volume	122	83	125	115	94	539
Accuracy	93.44 %	92.77 %	93.60 %	90.43 %	94.68 %	92.95 %

and integrity, while ensuring spatial alignment of this data. (2) Based on the multi-source data, three distinct virtual models of the geological conditions ahead of the tunnel face are constructed, providing a comprehensive mapping of the surrounding rock's physical properties from three different perspectives. (3) The proposed fuzzy cloud probability model effectively handles the uncertainty of geological conditions in complex environments, significantly improving tunnel construction safety and efficiency through intelligent rock classification. (4) The digital twin method proposed in this paper integrates multi-source information fusion, accurate 3D virtual model construction, and quantitative uncertainty decision at its technical core, significantly surpassing traditional approaches that rely solely on single data sources and expert judgment. Practical applications have demonstrated that this method offers clear advantages in enhancing prediction accuracy and construction responsiveness, providing robust decision support for safe tunnel construction in complex geological environments.

The future development of the advanced geological digital twin method can be further enhanced by expanding and refining the geological database. One promising direction is the integration of more data sources, such as combining real-time drilling data with GPR results. This integration aims to create a more comprehensive rock integrity database, addressing the limitation of GPR, which only reflects the integrity of a single cross-sectional plane of the surrounding rock. By merging different data sources, a more comprehensive representation of the surrounding rock's integrity can be achieved, especially its longitudinal variations. In addition, expanding the types of geological information in the database is also crucial. In this paper, the databases are constructed based on water content, lithology, and rock integrity. However, future research could explore additional dimensions, such as initial in-situ stress distribution and the orientation of structural planes, to construct more databases, further enriching the understanding of surrounding rock conditions. These new databases will help capture more diverse characteristics of the geological environment, improving the accuracy and practicality of digital twin models. Increasing the volume of data within existing databases is another key direction. Currently, the database size is limited by the range of geological information collected. Future studies could involve acquiring more geological data through advanced data collection technologies. As the database expands, more sophisticated artificial intelligence algorithms could be applied to identify patterns and correlations within the geological data, enabling the construction of a more precise and multi-dimensional geological digital twin model. With the application of more tunnel projects with various geological types on the iS3-based tunnel intelligent construction platform, the applicability of the method developed in this paper in different geological environments will be further explored, and the framework structure will be adjusted based on actual conditions. At

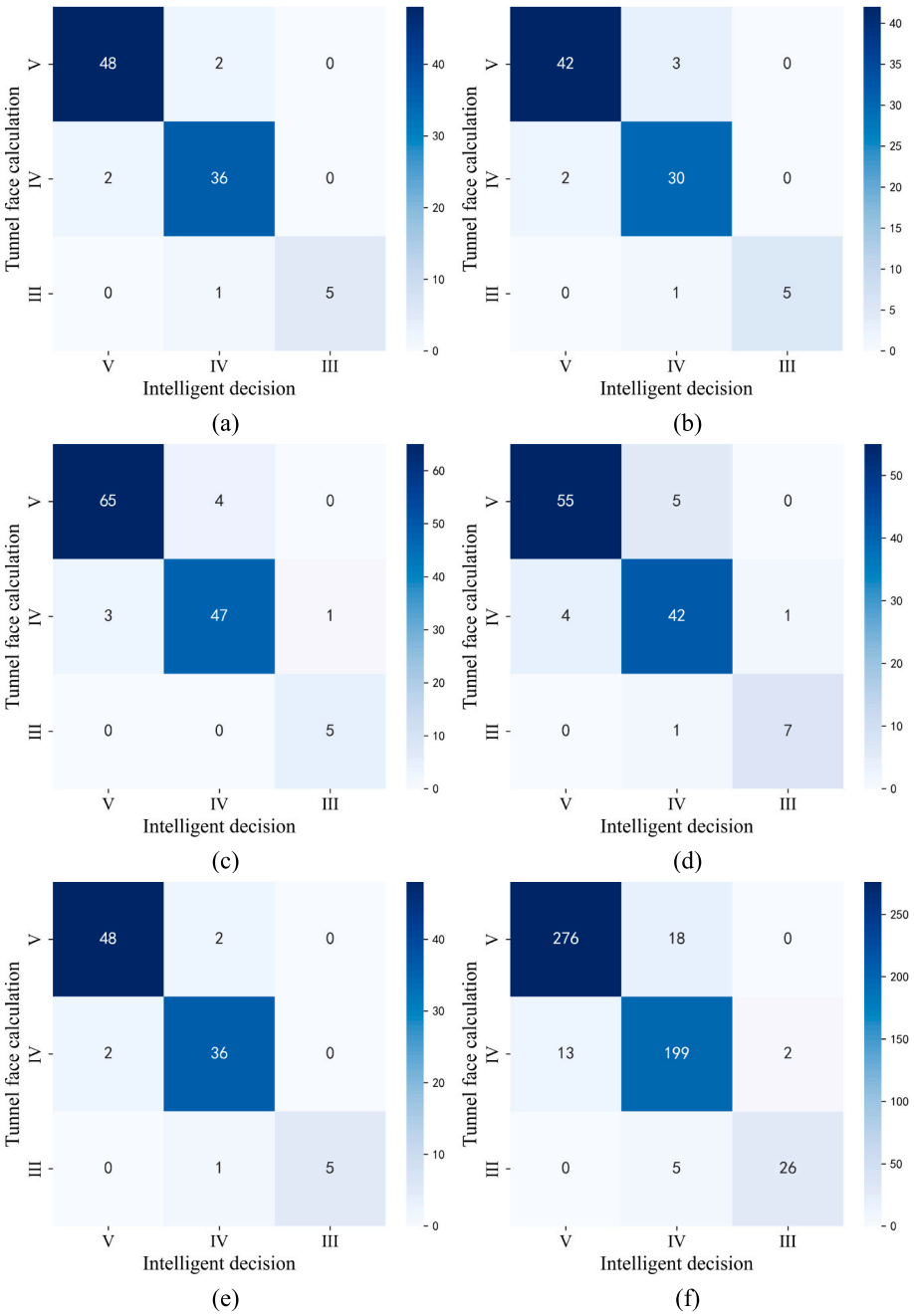


Fig. 22. Reliability verification confusion matrix; (a) Yanjiazhai tunnel; (b) Wuji tunnel; (c) Yongshan tunnel; (d) Chaoyang tunnel; (e) Yangjiawan tunnel; (f) The total sum of data from five tunnels.

the same time, it is hoped that more scholars will build upon this study to improve and develop digital twin methods suitable for different geological types.

CRediT authorship contribution statement

Huaiyuan Sun: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Data curation, Conceptualization. **Mengqi Zhu:** Writing – review & editing, Methodology, Conceptualization. **Yiming Dai:** Visualization, Software. **Xiaojun Li:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Yi Rui:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Weijia Zhu:** Resources, Data curation. **Hehua Zhu:** Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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